



*The entire world and its ecology
is your precious possession*

'LIVE AND LET LIVE'

Unit

8

Ecology and Environment

The 'Address and 'Profession' of organisms - A holistic approach to the study of organisms

The branches of Biology such as 'morphology' and 'anatomy' give us a fairly good picture of organisms. However **Modern Biology** lays stress on studying the habitat, environment - organism relationship (**Ecology**) to get a more comprehensive idea of life. Ecology deals with the study of organisms and their interrelationships (biotic factors) and interactions with the surrounding environment (abiotic factors). Interactions between the organisms and the environment limit the distribution of life. The study of ecology encompasses different levels – organism, population, community, ecosystem, etc.,. An important feature that may not be easily noticeable to the eye of an observer is – the 'steady state' an organism maintains (homeostasis), with reference to its surrounding abiotic factors. This type of homeostasis is achieved by morphological, physiological and behavioural adjustments / mechanisms. Whichever organism is adapted better to its environment 'on a long term basis' is offered '**POSITIVE NATURAL SELECTION PRESSURE**'. Thus ecological adaptations play an important role in '**SPECIATION**'. For example animals such as polar bears grow thicker coat of 'fur' during winter to insulate the body from the cooler environment. Some insects add 'glycerol' (anti-freezing agent) to their haemolymph to avoid freezing in winter. Certain animals show behavioural adaptations to survive – e.g. lizards 'bask' in the sun to warm up their body when sunlight is available and enter shade / burrows when it is very hot outside. Many cold blooded animals are '**temperature conformers**' and they change their temperature depending on that of the surroundings. Homeostasis of water (because water is the most important

constituent of protoplasm) is exhibited by all organisms including the microscopic **protists** and **bacteria**.

An eminent animal ecologist, **ODUM**, used the **analogy** to explain an organism's place and its function thus- IF AN ORGANISM'S HABITAT (place where it lives) IS ITS '**ADDRESS**', ITS ECOLOGICAL ROLE/FUNCTION (also called '**niche**') IS THE '**PROFESSION**' OF THE ORGANISM. Underlying all these aspects there are two things that are critically essential to support life on the Earth –

- a) Bio-geochemical Cycles involve uptake of various nutrients /minerals into the body, grow and ultimately return them to the environment to be '*recycled*' by other organisms, again and again. Thus 'life' comes from nature and goes back to nature – **NATURE'S FUNDAMENTAL PRINCIPLE**, so to say.
- b) The flow of 'ENERGY', mostly from the Sun, from one type of organism to another, 'uni-directionally', also has a major role to play in the LIFE on the Earth. It is important to 'preserve nature' without disturbing it too much, in the true spirit of by 'THINK GLOBALLY AND ACT LOCALLY' – a 'call' to the human race, given by the environmentalists in the **EARTH SUMMIT** of **Rio de Janeiro** - United Nations Conference on Environment and Development (**UNCED**).

There is an inbuilt '**hierarchy**' in organisms with reference to other neighbouring organisms and the environment in which they live – **individuals** – **populations** – **communities** – **ecosystems** – **biosphere** (global ecosystem). Mutations, gene recombinations, Natural selection process, etc., have important roles in the origin and distribution of organisms. With the origin of modern man, dispersal of organisms by man himself has become a part of this entire ecological world – the so called '**Human Effect**'. Exploring the ecology of organisms is a multidimensional, multidisciplinary task. It involves the study of 'population dynamics' and various allied factors such as natality, mortality, competition, predation, disease, space, food, shelter, influencing survival and propagation. Hunting, habitat destruction, overexploitation, modification of the environment by pollutants (**anthropogenic**: meaning '**man made**'), etc., are going to play a vital role in deciding the very existence of man on this planet. Let us not neglect the '**Greenhouse Effect**' and the related '**Global Warming**', destruction of the **ozone layer** in the stratosphere due to excessive use of chlorofluorocarbons (**CFCs**) – each one is going to affect **life** in its own way gradually diminishing chances of survival. Do your part to save mankind from the brink of disaster.

Lastly - **can you guess what your address and profession are ?** Your address is the 'terrestrial environment' and your profession is, you are a '**CONSUMER**' (Primary or secondary) and you are **perhaps** turning out to be '**self destroyer**' due to rapid urbanisation and 'conspicuous consumption' (over consumption).

Ecology and its Importance

8.0.1 What is ecology

8.0.2 Importance of Ecology

Our living world is fascinatingly diverse and amazingly complex. We can try to understand its complexity by investigating processes at various levels of biological organization -macromolecules, cells, tissues, organs, individual organisms, population, communities and ecosystems and biomes.

8.0.1 What is Ecology?

Ecology is a subject which deals with the study of the interactions among organisms and between the organisms and their physical (abiotic) environment. The word 'ecology' was derived from the Greek term 'oikos' which means 'house', and 'logos' meaning 'study'. Ecology was defined by **Ernst Haeckel** as "**the study of the relationship of organisms with their environment**". The environmental science, ecology, has two main branches- **autecology**, and **synecology**. Autecology is the ecology of a single species / population in relation to its environment. It is also known as '**species** (population) **ecology**'. Species ecology deals with the dynamics of 'species populations' and how these populations interact with their surrounding environment. It also deals with the study of changes in population sizes (**population dynamics**), over a period of time. Synecology is a branch of ecology that deals with the structure, development, and distribution of '**ecological communities**'. A **community**, as you perhaps know, is a group of organisms of different species living together and interacting with the surrounding environment. Synecology / community ecology deals with the interrelationships between organisms of different species living in a specified area and their interactions with their surrounding environment. It takes into consideration the functional roles of different organisms of the community and also the **community dynamics**.

8.0.2 Importance of Ecology

Civilization initiated with the use of fire and making of tools which led to the modifications in the environment. Gradually the technological advancement changed the life styles and human beings totally forgot their responsibilities to their surroundings. The negligence or ignorance towards 'natural resources' and the

‘**mother nature**’, gradually led to the disintegration of nature. To overcome this problem, it is time that humans act and save the nature for the future generations. The knowledge of organisms, environment, and the interrelationships is vital for the conservation of the Nature.

8.1 ORGANISMS AND ENVIRONMENT

8.1.1 Levels of organization

8.1.2 Environment

8.1.3 Habitat

8.1.4 Niche

Ecology at the organism level is essentially **physiological ecology** which tries to understand how different organisms are adapted to their environments in terms of not only survival but also reproduction. You may have learnt in earlier classes how the rotation of our planet around the

Sun and the tilt of its axis cause annual variations in the intensity and duration of light and temperature, resulting in the occurrence of distinct seasons. These variations together with annual variations in precipitation (rainfall) account for the formation of the major ‘**biomes**’ such as **desert**, **rain forest** and **tundra**.

Regional and local variations within each biome lead to the formation of a wide variety of habitats. On the planet Earth, life exists not just in a few favourable habitats only but even in extreme and harsh habitats such as the hot Rajasthan desert, rain-soaked Meghalaya forests, deep ocean trenches, fast flowing streams, polar regions, high mountain tops, hot thermal springs, and stinking ‘compost pits’, to name a few. Even our intestine is a **unique habitat** for hundreds of species of microbes.

What are the key elements that lead to so much variation in the physical and chemical conditions of different habitats?

The most important ones are temperature, water, light and soil. We must remember that the physico-chemical (**abiotic**) components alone do not characterize the habitat of an organism completely; the habitat includes **biotic components** also - pathogens, parasites, predators and competitors - of the organism with which they interact constantly. We assume that over a period of time, organisms evolved adaptations for survival and reproduction, in their habitats through **Natural Selection**.

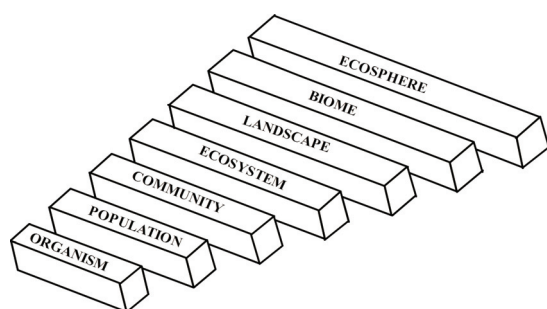


Fig. 8.1 Levels of organization

8.1.1 Levels of organization

Ecology is basically concerned with four levels of biological organization-**organisms**, **populations**, **communities** and **biomes**. In this chapter we explore ecology at organism and population levels.

Ecological Hierarchy means arrangement into a '**graded series**'. Ecological organization consists of eleven integrative levels, ranging from Cell to Ecosphere - cell, tissue, organ, organ- system, organism, population, community, ecosystem, landscape, biome and ecosphere (also called Biosphere).

1. Population

Population is a group of organisms of the same species, living in a specific area at a specific time. For example, the fish belonging to the species *Catla catla* living at a given time, in a pond constitute a 'population'. In the same way, the people (*Homo sapiens*) of India in 2012 form the **Indian population**. Populations are regulated by a set of factors such as **natality**, **mortality**, and **population density**.

2. Community

It is an association of the interacting members of populations of different autotrophic and heterotrophic species in a particular area. In a community, generally one or a few species dominate with reference to their numbers or size. Generally, a community is named after its dominant population. For example, the Himalayan region (at the altitude of 5000 to 8000 ft.) is dominated by the pine group plants – **deodar trees**. So, it is described as the **Pine-Deodar community**.



Fig. 8.2. Pine-deodar community

3. Ecosystem

It is the next level of organization above the level of *biological community*. An **ecosystem** is a functional unit of the biosphere in which members of the community interact among themselves and with the surrounding environment, involving '**flow of energy**' forming a well defined **trophic structure**. An ecosystem can be as small as an aquarium or a tiny puddle. Ecosystems sometimes may overlap, as the boundaries of an ecosystem are flexible.

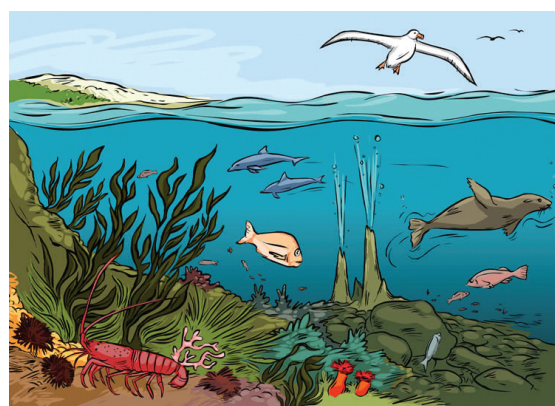


Fig. 8.3 Ecosystem

4. Landscape

It is the unit of land containing different ecosystems (mosaic of ecosystems) surrounded by natural boundaries. It is the level of organization higher than 'ecosystem'.

5. Biome

A 'biome' is a **large community of plants and animals** that occupies a vast **region**. There are 'terrestrial biomes' and 'aquatic biomes'. Terrestrial biomes are characterized by **specific climate** and **dominant vegetation** e.g. tropical rain forest, desert, coniferous forest, tundra etc. Aquatic biomes include freshwater biomes and marine biomes.

6. Ecosphere (Biosphere)

All the habitable zones on the Earth constitute the ecosphere or biosphere. It is the part of the Earth that supports 'life'. It extends several kilometers above the Earth's surface into the atmosphere and extends several kilometers below the ocean's surface. The biosphere comprises all of the Earth's biomes.

8.1.2 Environment

Environment is the sum total of biotic and abiotic factors present around the organisms influencing them in various ways. Any component of the environment that influences the organism is called **environmental factor**. The environmental factors are of three types i) climatic, ii) topographical, iii) edaphic.

- i) Climate refers to average weather conditions like temperature, humidity, cloud cover, wind velocity, rainfall, etc.
- ii) Topography refers to the surface features (physical features such as elevation / altitude) of a place or region. The topography affects the climatic conditions of the region.
- iii) Edaphic factors are related to the soil, and they influence the life of organisms. The main edaphic factors include soil texture, pH, content of organic matter, water, etc.

8.1.3 Habitat and Medium

Ecologically, **habitat** is the place in which an organism lives. It is comparable to the '**address**' of a person (as mentioned in the introduction page to ecology). For instance, the habitat of fish is a pond, lake, sea etc., the habitat of a lion is forest, the habitat of *Ascaris* is the 'small intestine' of man, and so on. The **water** surrounding the body of a fish is called the **medium** and the medium of a lion is the **air** around its body.

8.1.4 Niche

Within a community, each organism occupies a particular biological role or **niche**. Niche is the **functional role** of an organism in an ecosystem (comparable to the '**profession**' of a person, as mentioned in the introduction). It describes the relational position of a species in its ecosystem. For example, the ecological niche of a rose bush growing in the backyard includes absorbing light, water and nutrients for photosynthesis, releasing oxygen into the atmosphere, providing shelter and food for other organisms (e.g. **aphids**) – **in short the 'rose bush' is a 'producer'**.

8.2 ECOSYSTEM- ELEMENTARY ASPECTS

- 8.2.1 Light
- 8.2.2 Temperature
- 8.2.3 Water
- 8.2.4 Soil
- 8.2.5 Pressure

An 'ecosystem' consists of both **structural** components and **functional** components. The structural components of ecosystem are of two types – **abiotic** and **biotic** factors. Abiotic factors are again of two types- physical and chemical factors. The physical factors are light, temperature, soil, pressure, etc., Oxygen, carbon dioxide and the various mineral nutrients in the soil / water constitute the chemical factors. These physical and chemical factors affect the life of the organisms present in an ecosystem. Energy transfer and recycling of minerals constitute the functional aspect of an ecosystem.

8.2.1 Light

Since plants produce food through photosynthesis, a process which is possible only when sunlight is available as a source of energy, we can easily understand the importance of light for all living organisms, particularly the 'autotrophs'. Many species of small plants (herbs and shrubs) growing in forests are adapted to photosynthesize optimally under very low light conditions because they are constantly overshadowed by tall trees with well spread branches. Many plants are also dependent on sunlight to meet their **photoperiod requirement** for flowering.

For many animals, light is important in that they use the diurnal and seasonal variations in the **intensity** and **duration (photoperiod)** of light, as cues (indicators) for timing their foraging (collection of food), reproductive and migratory activities. The availability of light on land is closely linked with that of temperature since the sun is the source of both. But, deep in the oceans, the environment is always dark and its inhabitants do not experience light. The spectral quality of solar radiation is also important for life. The UV component of the spectrum is harmful to many organisms. All the colour components of the visible spectrum are not available for marine plants, living at different depths of the seas.

I. Biological effects of light

Light is an ecological factor that shows its influence on many biological phenomena and activities like pigmentation, movement, vision, metabolism, etc.,

1. Effect of light on pigmentation

Light influences the colour of the skin. The animals which live in regions of low intensity of light such as caves, have less pigmentation than that of the animals exposed to light. For example a cave dwelling amphibian, **Proteus anguinus**, has pale coloured skin because of less pigmentation. When it is brought into sunlight the skin colour turns *dark* over a period of time.

2. Effect of light on direction and rate of movement

Response shown by an organism to changes in light is called **photoresponse**. The major photoresponses of motile organisms include:

- i) **Phototaxis** is oriented locomotor movement of an organism towards or away from the direction of light as seen in euglenas(positive response), cockroaches(negative response) and,
- ii) **Photokinesis** is the influence of light on non- directional movement of organisms as seen in the larvae of *Pinnotheres maculatus*- the mussel crab, in which intensity of light influences the velocity of the movement of the organism.

Phototropism: The orientation to light of a non-motile organism such as a plant or plant part is called phototropism. For example the growth mediated bending of plants towards sunlight is due to phototropism.

3. Influence of Light on Vision and Behavior

Light is essential for vision, to procure food, self protection, identification and selection of the mate for sexual reproduction. It also influences the behaviour of organisms. Some animals are active during the 'day time' (**diurnal** animals e.g. majority of the birds, reptiles and mammals. Others, which are active during night time are called **nocturnal** animals e.g. earthworms, cockroaches, etc.,

4. Effect of light on metabolism

The intensity of light influences the rate of metabolism in animals. At lower intensities of light the rate of metabolism slows down, where as in higher intensities it increases.

5. Effect of light on biological rhythms

By definition rhythm is a change that is repeated with similar pattern in a time frame. In the bodies of organisms many behavioural activities are repeated at regular intervals and these are called 'biological rhythms'.

Biological rhythms that occur in a time period of 24 hours are **circadian rhythms**. If the biological rhythms repeat annually (every year), they are called **circannual rhythms**. They are controlled by '**biological clocks**' present in many living organisms. Light has a role in setting or resetting the biological clocks.

6. Effect of 'photoperiod' on animals (photoperiodism)

The duration of the light hours / exposure to light in a day is known as **photoperiod**. The response of organisms for the photoperiod is called **photoperiodism**. The response to photoperiod often centers on reproduction and survival(production of flowers, the migration of birds etc.,). The specific day length which is essential for the

initiation of seasonal events is called '**critical photoperiod**'. The length of critical photoperiod is not the same in all the species, it varies from species to species and also within the same species inhabiting at various latitudes. For example, when winter sets in Siberia, the light available time period in a day decreases, and so some birds migrate all the way to the new **feeding** and **breeding grounds** in various parts of India. When it is summer time back in Siberia, they migrate back to their homeland.

Exposure of young fowls in poultry to more lighted hours in a day induces speedy maturity and the egg laying process. It is a familiar example to most of us.

Bioluminescence: Production of light by certain living organisms is called bioluminescence. The light emitted by living organisms is devoid of infrared rays, and so it is called '*cold light*'. Some Jelly fishes (cnidarian), *Chaetopterus* (annelid), firefly (arthropod), squids (mollusc), *Pyrosoma* (protochordate), and some fishes produce bioluminescence. It is believed to *influence life* in the dark habitat by providing intra-specific communication, sexual attraction, attracting the prey, sending protective warning signals, etc.

II. Effect of UV Rays

The short wave lengths of light ranging from 100 nm to 380nm are called **Ultraviolet rays**. UV radiation is classified into three categories: UVC light (100nm to 280 nm), UVB (280nm to 320nm) and UVA (320nm to 380nm). UV radiation kills the microorganisms present on the body surfaces of animals. UV radiation helps in the conversion of sterols present in the skin into vitamin D in mammals. Compared to UVA rays, UVB and UVC rays are more harmful to organisms.

8.2.2 Temperature

Temperature is a measure of the intensity of heat. The temperature on land or in water is not uniform. On land the temperature variations are more pronounced when compared to the aquatic medium, because land absorbs or loses heat much quicker than water. The temperature on land depends on seasons and the geographical area on this planet. Temperature decreases progressively when we move from the equator to the poles. Altitude also causes variations in temperature. For instance, the temperature decreases gradually as we move to the top of the mountains.

I. Thermal Stratification

Temperature variations occur with seasonal changes in the temperate regions. These differences in the temperature form 'thermal layers' in water. These phenomena are called **thermal stratifications**.

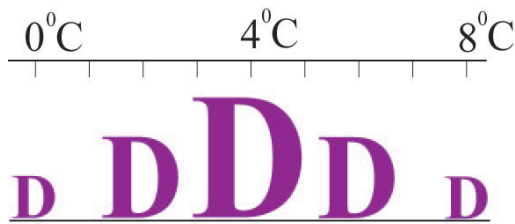


Fig. 8.4 Density decreases above or below 4°C

Water shows maximum density at 4°C. Rise or fall of temperatures above or below 4°C decreases its density. This **anomalous property** of water and the seasonal variations in temperature are responsible for the thermal stratification in temperate lakes.

Summer stratification

During summer in temperate lakes, the density of the surface water decreases because of increase in its temperature (21-25°C). This 'upper more warm layer' of a lake is called epilimnion. Below the epilimnion there is a zone in which the temperature decreases at the rate of 1°C per meter in depth, and it is called **thermocline** or **metalimnion**. The bottom layer is the **hypolimnion**, where water is relatively cool, stagnant and with low oxygen content (*due to absence of photosynthetic activity*).

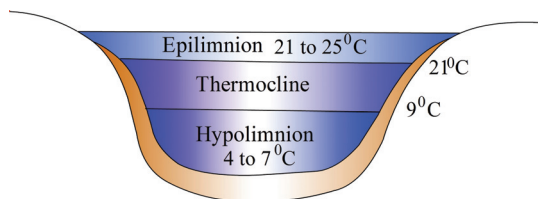


Fig. 8.5 Summer stratification in a lake

During **autumn** (also called **Fall**), the epilimnion cools down, and the surface water becomes heavy when the temperature is 4°C, and sinks to the bottom of the lake. Overturns bring about 'uniform temperature' in lakes during that period. This circulation during the autumn is known as the **fall overturn** or **autumn overturn**. The upper oxygen rich

water reaches the hypolimnion and the nutrient rich bottom water comes to the surface. Thus there is uniform distribution of nutrients and oxygen in the lake.

Winter stratification /stagnation

The 'Fall' is followed by 'Winter'. In this season the surface water cools down. The upper water freezes when the temperature reaches 0°C. Below the upper icy layer, the

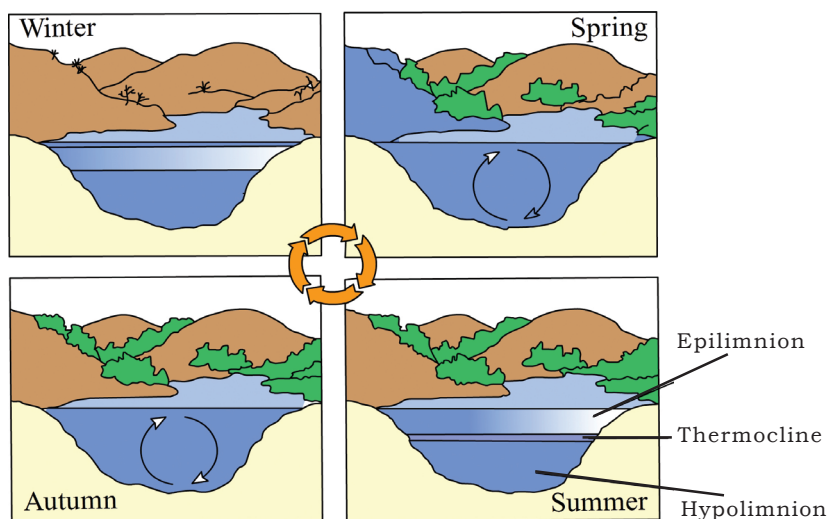


Fig. 8.6 Stratification in temperate lakes

cool (4°C) water occupies the lake. The aquatic animals continue their life below the icy layer. At lower temperatures the activity of bacteria and the rate of oxygen consumption by aquatic animals decrease. Hence, organisms can survive below the frozen upper water without being subjected to '**hypoxia**' (low oxygen availability). In the 'Spring season' the temperatures start rising. When it reaches 4°C, the

water becomes more dense and heavy and sinks to the bottom, taking 'oxygen rich water' to the bottom. The upper oxygen rich water sinks down and the bottom 'nutrient rich water' reaches the surface. It is called '**spring overturn**'. The lakes which show overturns twice a year are called '**dimictic lakes**'. Thus 'stratifications' and 'overturns' help survival of organisms at all levels in deep lakes, as they help in redistribution of oxygen.

II. Biological effects of Temperature

1. Temperature Tolerance

A few organisms can tolerate and thrive in a wide range of temperatures they are called **eurythermal** (e.g. all homoeotherms), but, a vast majority of organisms are restricted to a narrow range of temperatures (such organisms are called **stenothermal** e.g. fishes and coral animals). The levels of thermal tolerance of different species determine their geographical distribution.

2. Temperature and Metabolism

Temperature affects the *working of enzymes* and through it, the **basal metabolism**, and other physiological functions of organism. The temperature at which the metabolic activities occur at the climax level is called the '**optimum temperature**'. The lowest temperature at which an organism can live indefinitely is called **minimum effective temperature**. If an animal or plant is subjected to a temperature below the minimum effective limit, it enters into a condition of inactiveness called **chill coma**. The metabolic rate increases with the rise of temperature from the **minimum effective temperature** to **optimum temperature**.

The maximum temperature at which a species can live indefinitely in an active state is called **maximum effective temperature**. If the temperature is raised above the maximum effective temperature, the animals enter into '**heat coma**'. The maximum temperature varies much in different animals.

3. van't Hoff's rule

van't Hoff, a Nobel Laureate in thermochemistry, proposed that, with the increase of every 10°C, the rate of metabolic activities doubles. This rule is referred to as the **van't Hoff's rule**. van't Hoff's rule can also be stated in reverse saying that the reaction rate is halved with the decrease of every 10°C. The effect of temperature on the rate of a reaction is expressed in terms of **temperature coefficient** or Q_{10} value. Q_{10} values are estimated taking the ratio between **the rate of a reaction at X°C** and **rate of reaction at (X-10°C)**. In the 'living systems' the Q_{10} value is about 2.0. If the Q_{10} value is 2.0, it means, for every 10°C increase, the rate of metabolism doubles.

4. Cyclomorphosis

The cyclic seasonal morphological variations among certain organisms is called **cyclomorphosis**. This phenomenon has been demonstrated in the cladoceran (a sub group of crustacea) *Daphnia* (*water flea*). In the winter season the head of *Daphnia* is

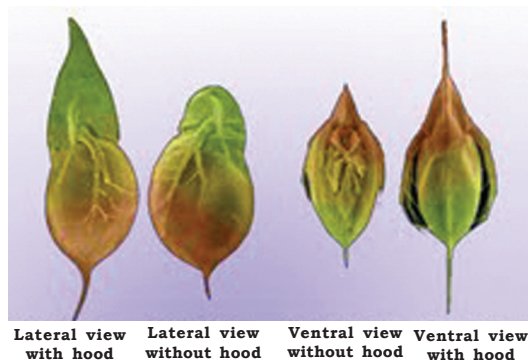


Fig. 8.7 Cyclomorphosis in *Daphnia*

‘round’ in shape (typical or non-helmet morph). With the onset of the spring season, a small ‘helmet’/‘hood’ starts developing on it. The helmet attains the maximum size in summer. In ‘autumn’ the helmet starts receding. By the winter season, the head becomes round. Some scientists are of the opinion that **cyclomorphosis** is a seasonal adaptation to changing densities of the water in lakes – in summer as the water is less dense *Daphnia* requires a larger body surface to keep floating easily. During winter the water is more dense, and so it does not require a larger surface area of the body to keep floating. Others

believe that these cyclic changes are adaptations to ‘**stabilize the movement**’ in water. Compared to the ‘typical morphs’, the ‘helmeted morphs’ can **resist the water currents better** to stay in the water rich in food materials.

5. Temperature adaptations

Temperature adaptations in animals can be dealt under three heads a) Behavioural adaptations, b) Morphological and Anatomical adaptations and c) Physiological adaptations.

a. Behavioural adaptations

Some organisms show behavioural responses to cope with variations in their environment. **Desert lizards** manage to keep their body temperature fairly constant by behavioural means. They ‘bask’ (staying in the warmth of sunlight) in the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the temperature starts increasing. Some species are capable of burrowing into the soil to escape from the excessive heat above the ground level.

b. Morphological and anatomical adaptations

In the polar seas, aquatic mammals such as the seals have a thick layer of fat (**blubber**) that acts as an **insulator** and reduces the loss of body heat, underneath their skin.

The animals which live in colder regions have larger body size with greater mass. The body mass is useful to generate more heat. As per **Bergmann’s rule** mammals and other warm blooded animals living in colder regions have less ‘**surface area to body volume ratio**’, than their counterparts living in the tropical regions.

The small surface area helps to conserve heat by reducing the surface area of heat loss. For instance, the body size of American moose / Eurasian elk (*Alces alces*), increases with the latitudes in which they live. ^[1] Moose of northern part of Sweden shows 15-20% more **body mass** than the same species (counterparts) living in the southern Sweden.

Mammals from colder climates generally have shorter earlobes and limbs (extremities of the body) to minimize heat loss. Large earlobes and long limbs increase the surface area without changing the body volume. This is known as **Allen's rule**. For instance, the polar fox, *Vulpes lagopus* (formerly called *Alopex lagopus*), has short extremities to minimize the heat loss from the body. In contrast, the desert fox, *Vulpes zerda*, has large ear lobes and limbs to facilitate better 'heat loss' from the body.

c. Physiological adaptations

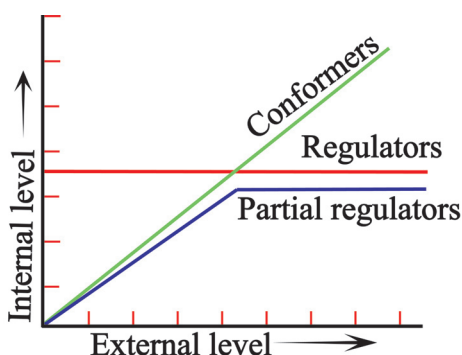
In most animals, all the physiological functions proceed 'optimally' in a narrow temperature range (in humans, it is 37°C). However there are microbes (**archaebacteria**) that flourish in hot springs and in some parts of deep seas, where temperatures far exceed 100°C. Many fish thrive in Antarctic waters where the temperature is always below zero. Having realized that the abiotic conditions of many habitats may vary over a time period, we now ask - **How do the organisms living in such habitats manage with stressful conditions?** One would expect that during the course of millions of years of their existence, many species would have evolved a relatively constant internal (within the body) environment. It permits all biochemical reactions and physiological functions to proceed with maximal efficiency and thus, enhance the overall 'fitness' of the species. This constancy, could be chiefly in terms of **optimal temperature** and **optimal osmotic concentration** of body fluids. So the organism should try to maintain the constancy of its internal environment (homeostasis) despite varying external environmental conditions that tend to upset its homeostasis. This is achieved by the processes described below.

(i) Regulate

Some organisms are able to maintain homeostasis by physiological (sometimes behavioural also) means which ensures constant body temperature, constant osmotic concentration, etc., All birds and mammals, and a very few lower vertebrate and invertebrate species are indeed capable of such regulation (**thermoregulation** and **osmoregulation**). Evolutionary biologists believe that the 'success' of mammals is largely due to their ability to maintain a constant body temperature and thrive whether they live in Antarctica or in the Sahara desert.

The mechanisms used by most mammals to regulate their body temperature are similar to the ones that we, the humans use. We maintain a constant body

[1] The traditional example of comparison between the large sized polar bear and the small sized tropical black bear to support Bergmann's rule, is not acceptable to some Modern Ecologists as these bears belong to two different Genera, altogether (i.e., not the same species to be compared).



temperature of 37°C . In summer, when the outside temperature is more than our body temperature, we sweat profusely. The resulting '**evaporative cooling**' brings down the body temperature. In winter when the temperature is much lower than 37°C , we start to **shiver** (a kind of exercise which produces heat and raises the body temperature – a type of body's own defence mechanism against low temperatures). Plants, on the other hand, do not have such mechanisms to maintain internal temperatures.

(ii) Conform^[2]

Majority (99 percent) of animals cannot maintain a constant internal environment. Their body temperature changes with the ambient (surrounding) temperature. In aquatic animals, the osmotic concentration of the body fluids changes along with that of the surrounding water. Such animals are described as '**conformers**'.

(iii) Partially regulate^[3]

Animals such as 'camels' can be 'conformers' up to a particular range of temperature and 'regulator' afterwards. So, they are described as 'partial regulators' or 'partial conformers'.

Why did not the conformers evolve to become regulators?

Thermoregulation is energetically 'expensive' for many organisms. This is particularly true in small animals like shrews and humming birds. Heat loss or heat gain is a function of the surface area. Since small animals have a larger surface area relative to their volume, they tend to lose body heat very fast when it is cold outside; then they have to spend much energy to generate body heat through metabolism. This is the main reason why very small animals are rarely found in the 'polar regions'. During the course of evolution, the costs and benefits of maintaining a constant internal environment are taken into consideration. Some species have evolved the ability to regulate, but only over a limited range of environmental conditions, beyond which they simply conform.

If the stressful external conditions are localized or remain only for a short duration, the organism has two other alternatives.

(iv) Migrate

The organism can move away temporarily from the 'stressful habitat' to a more 'hospitable' (comfortable) area and return when the stressful period is over. In human

^[2] Adapt from one condition to a new or different conditions.

^[3] There are no perfect regulators or perfect conformers. There are different intermediate grades among animals.

analogy (comparison), this strategy is comparable a person moving from Delhi to Shimla for the duration of summer. Many animals, particularly birds, during winter undertake long-distance migrations to more hospitable areas. Every winter, many places in India including the famous Keoladeo or **Keoladeo Ghana National Park** (Formerly - Bharatpur bird sanctuary) in Rajasthan and Pulicat Lake in Andhra Pradesh host thousands of 'migratory birds' coming from Siberia and other extremely cold northern regions.

(v) Suspend life activities

In bacteria, fungi and lower plants, various kinds of thick-walled spores are formed which help them survive unfavourable conditions. They germinate (come out of the spore wall and produce a normal active organism) on the return of suitable environmental conditions.

Plant Examples : In higher plants, seeds and some other vegetative reproductive structures serve as means **to tide over periods of stress** besides helping in dispersal - they germinate to form new plants under favourable moisture and temperature conditions. They do so by reducing their metabolic activity and going into a state of 'dormancy'.

Some animals can avoid the stress by escaping in 'time' (migration is- escaping in 'space'). The familiar case of 'polar bears' going into **hibernation** during winter is an example of **escape in time**. Some snails, fish and frogs go into **aestivation** to avoid summer-related problems-heat and desiccation.

6. Diapause

Certain organisms show delay in development, during periods of unfavourable environmental conditions and spend some period in a state of 'inactiveness' called '**diapause**'. This dormant period in animals is a mechanism to survive extremes of temperature, drought, etc. It is seen mostly in insects and embryos of some fish. Under unfavourable conditions many zooplankton species in lakes and ponds are known to enter diapause (Ref:NCERT Text Book).

8.2.3 Water

Water is another important factor influencing the life of organisms. Life is unsustainable without water. Its availability is so limited in deserts that only certain **special adaptations** make it possible for them to live there. You might think that organisms living in oceans, lakes and rivers should not face any water-related problems, but it is not true. For aquatic organisms the quality (chemical composition, pH, etc.) of water becomes important. **The salt concentration (measured as salinity in parts per 1000) is less than 5 in inland waters, and 30-35 in the seawater.** Some organisms are tolerant to a wide range of salinities (**euryhaline**), but others are restricted to a narrow range (**stenohaline**). Many freshwater animals cannot live for long in sea water and vice versa because of the osmotic problems, they would face.

1. Adaptations in freshwater habitat

Animals living in freshwaters have to tackle the problem of **endosmosis**. The osmotic pressure of freshwater is very low and that of the body fluids of freshwater organisms is much higher. So water tends to enter into bodies by endosmosis. To maintain the balance of water in the bodies, the freshwater organisms acquired several adaptations such as, contractile vacuoles in the freshwater protozoans, large **glomerular kidneys** in fishes, etc., They send out large quantities of urine, along which some salts are also lost. To compensate the 'salt loss' through urine, freshwater fishes have 'salt absorbing' '**chloride cells**' in their gills. The major problem in freshwater ponds is - in summer most of the ponds dry up. To overcome this problem most of the freshwater protists undergo **encystment**. The freshwater sponges produce asexual reproductive bodies, called **gemmules**, to tide over the unfavourable conditions of the summer. The 'African lungfish', *Protopterus*, burrows into the mud and forms a 'gelatinous cocoon' around it, to survive, in summer.

2. Adaptations in marine habitat

Seawater is high in salt content compared to that of the body fluids. So, the marine animals continuously tend to lose water from their bodies by **exosmosis** and face the problem of dehydration. To overcome the problem of water loss, marine fishes have **agglomerular kidneys** with less number of nephrons. Such kidneys minimize the loss of water through urine. To compensate water loss the marine fish drink more water, and along with this water, salts are added to the body fluids and disturb the internal equilibrium. To maintain salt balance (**salt homeostasis**) in the body, they have **salt secreting chloride cells** in their **gills**. Marine birds like **sea gulls** and **penguins** eliminate salts in the form of salty fluid that drips through their nostrils. In turtles the ducts of chloride secreting glands open near the eyes. Some cartilaginous fishes retain **urea** and **trimethylamine oxide** (TMO) in their blood to keep the body fluid **isotonic** to the sea water and avoid dehydration of the body due to exosmosis.

3. Water related adaptations in brackish water animals

The animals of brackish water are adapted to withstand wide fluctuations in salinity. Such organisms are called **euryhaline** animals. The migratory fishes such as **salmon** and **Hilsa** are **anadromous** fishes i.e., they migrate from the sea to freshwater, for breeding ; *Anguilla bengalensis* is a **catadromous** fish i.e., it migrates from the river to sea, for breeding. In these fishes their **glomerular kidneys** are adjusted to changing salinities. The chloride cells are adapted to **excrete** or **absorb** salts depending on the situation. On entering the river salmon drinks more freshwater to maintain the concentration of body fluids equal to that of the surround water.

4. Water related adaptations for terrestrial life

In the absence of an external source of water, the **kangaroo rat** of the North American deserts is capable of meeting all its water requirements through oxidation of its internal fat (in which water is a by product-*metabolic water*). It also has the ability to concentrate its urine, so that minimal volume of water is lost in the process of removal of their excretory products.

Plant Example: Many desert plants have a thick cuticle on their leaf surfaces and have their stomata arranged in deep pits to minimize water loss through transpiration. They also have a special photosynthetic pathway (⁴CAM) that enables their stomata to remain closed during day time. Some desert plants like *Opuntia*, have no leaves - they are reduced to spines -and the photosynthetic function is taken over by the flattened stems to reduce loss of water (Ref: NCERT Text BookK).

8.2.4 Soil

The nature and properties of soil in different places vary depending on the climate, and the '*weathering*⁵' processes involved. Various characteristics of the soil such as soil composition, grain size and aggregation determine the percolation and water holding capacity of the soils. These characteristics, along with the parameters such as pH, mineral composition etc., determine to a large extent the vegetation in any area. This in turn dictates the type of animals that can be supported. Similarly, in the aquatic environment, the sediment-characteristics often determine the type of benthic animals that can live there.

8.2.5 Pressure

Pressure is another factor that changes dramatically with depth in the ocean. Organisms on land face less than one 'atmosphere' of pressure at the sea level. Since water is much heavier than air, marine organisms are under much more pressure than those on land. The pressure in water increases at the rate of 1 atmosphere per 10m depth. The organisms living in such extreme environments show a wide range of biochemical adaptations. Some organisms possess adaptations that are physiological and allow them to respond quickly to a stressful situation. If you had ever been to any high altitude place (e.g. >3,500m **Rohtang Pass** near **Manali** and **Manasarovar**, in Tibet) you must have experienced what is called **altitude sickness**. Its symptoms include **nausea** (vomiting sense), **fatigue** (tiredness) and **heart palpitations** (abnormality in heart beat). This is because in the low atmospheric pressure of high altitudes, the body does not get enough oxygen. But, you gradually get acclimatized and overcome the altitude sickness. How did your body solve this problem? The body compensates low oxygen availability by increasing red blood cell production and increasing the rate of breathing. (Note: **decreasing the binding capacity of haemoglobin** as mentioned in *NCERT Text Book, Page-226*).

[⁴] Crassulacean Acid Metabolism Pathway.

[⁵] Weathering is a collective term for the processes by which rock at or near the earth's surface is disintegrated and decomposed by the action of wind, water etc., to form soil.

8.3 POPULATION INTERACTIONS

8.3.1 Inter-specific Interactions

8.3.2 Predation

8.3.3 Competition

8.3.4 Parasitism

8.3.5 Commensalism

8.3.6 Mutualism

Can you think of any natural habitat on earth that is inhabited just by a single species? There is no such habitat and such a situation is unimaginable. For any species, the minimal requirement is one more species on which it can feed. Even a plant species, which makes its own food, cannot survive alone; it needs soil microbes to break down the organic matter in soil and return the inorganic nutrients for absorption. And

then, how will the plant manage pollination without an animal agent? It is obvious that in nature, animals, plants and microbes do not and cannot live in isolation but interact in various ways to form a **biological community**. Even in minimal communities, many interactive linkages exist, although all may not be readily visible.

8.3.1 Inter-specific Interactions

Inter-specific interactions arise from the interaction of populations of two different species. They could be beneficial, detrimental or neutral (neither harmful nor beneficial) to one of the species or both. Assigning a '+' sign for beneficial interaction, '-' sign for detrimental and '0' for neutral interaction, let us look at all the possible outcomes of inter-specific interactions.

The interactions between species are grouped into six types. They are mutualism, commensalism, parasitism and amensalism. Both the species benefit in **mutualism** and both lose in **competition** in their interactions with each other. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called **commensalism**. In **amensalism** on the other hand one species is harmed whereas the other is unaffected. In both **parasitism** and **predation** only one species benefits (parasite and predator, respectively) and the interaction is detrimental to the other species (**host** and **prey**, respectively). Predation, parasitism and commensalisms share a common characteristic - the interacting species live closely together.

Table 8.1: Population Interactions-Types

Name of Interaction	Species A	Species B
Mutualism	+	+
Competition	-	-
Predation	+	-
Parasitism	+	-
Commensalism	+	0
Amensalism	-	0

[¶] Predation is not an association (it is a feeding strategy). It is an interaction between two different species. The predator gets benefit at the cost of the prey.

8.3.2 Predation

What would happen to all the energy fixed by autotrophic organisms if the community has no animals to eat the plants? You can consider predation as nature's way of transferring the energy fixed by plants to higher trophic levels. When we think of predator and prey, most probably it is the tiger and the deer that readily come to our mind, but a sparrow eating any seed is also a type of predator (a seed predator-also called **granivore**). Although animals eating plants are categorized separately as herbivores, they are, in a broad ecological context, not very different from predators.

Besides acting as 'conduits' / 'pipelines' for energy transfer across trophic levels, predators play other important roles. They keep the prey populations under control. In the absence of predators, the prey species could achieve very high population densities and cause instability in the ecosystem. Predators have different types of functions to play in nature. They include:

A. Predator as a biological control agent

The prickly pear cactus introduced into Australia in the early 1920s caused havoc by spreading rapidly into millions of hectares of rangeland (vast natural grass lands). Finally, the invasive cactus was brought under control only after a cactus-feeding predator (a moth) was introduced into the country. **Biological control** methods adopted in agricultural pest control are based on the ability of the predators to regulate prey populations.

B. Predators maintain 'species diversity'

Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species. In the rocky intertidal communities of the American Pacific Coast, the starfish *Pisaster* is an important predator. In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year in that experimental area, because of increased **inter - specific competition**.

C. Predators are prudent (practical) pertaining to preys

If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct due to lack of food. This is the reason why predators in nature are 'prudent'.

Prey species have evolved various defenses to lessen the impact of predation. They include :

- a) **Preys fool (deceive) or avoid their predators**: Some species of insects and frogs are cryptically-coloured (**camouflaged**) to avoid being detected easily by the predator. Some are poisonous and therefore avoided by the predators.
- b) **Preys defend by becoming distasteful to predators**: The **Monarch butterfly** is highly distasteful to its predator (bird) because of a special chemical present in its

body. Interestingly, the butterfly acquires this chemical during its **caterpillar stage** by feeding on a **poisonous weed**.

- c) Plants too have their defensive mechanisms:** For plants, herbivores are the predators. Nearly 25 per cent of all insects are known to be **phytophagous** (feeding on plant sap and other parts of plants). The problem is particularly severe for plants because, unlike animals, they cannot escape from their predators. Plants therefore have evolved a variety of morphological and chemical defences against herbivores.
- i) **Thorns** (Acacia, Cactus, etc.,) are the most common morphological means of defense. Many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it.
 - ii) You must have seen the weed Calotropis growing in abandoned fields. The plant produces highly poisonous **cardiac glycosides** and that is why you never see any cattle or goats browsing on this plant.
 - iii) A wide variety of chemical substances that we extract from plants on a commercial scale (**nicotine, caffeine, quinine, strychnine, opium**, etc.,) are produced by them actually as defences against grazers and browsers.

8.3.3 Competition

When Darwin spoke of the struggle for existence and survival of the fittest in nature, he was convinced that **interspecific competition** is a 'potent force' in the process of organic evolution, involving Natural Selection. It is generally believed that competition occurs when closely related species compete for the same resources that are limited, but this is not entirely true.

I. Competition among Unrelated Species

Firstly, unrelated species could also compete for the same resource (Interspecific competition). For instance, in some **shallow** South American lakes visiting **flamingos** and resident **fishes** compete for their common food, the suspended **zooplankton** in the shallow waters. Secondly, resources need not be limiting for competition to occur. In **interference competition**, the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species, even if resources (food and space) are abundant. Therefore, competition is best defined as a process in which the fitness of one species (measured in terms of its '**r**'- the intrinsic rate of increase) is significantly lower in the presence of another species.

II. Competitive exclusion

It is relatively easy to demonstrate in laboratory experiments, as **Gause** and other experimental ecologists did. When the resources are limited, the competitively superior species will eventually eliminate the other species e.g. the Abingdon tortoise in Galapagos

Islands became extinct within a decade after goats were introduced on the island, actually due to the greater browsing efficiency of the goats.

III. Competitive release

Another evidence for the occurrence of competition in nature comes from what is called '**competitive release**'. Competitive release occurs when one of the two competing species is removed from an area, thereby releasing the remaining species from one of the factors that limited its population size. A species, whose distribution is restricted to a small geographical area because of the presence of a competitively superior species, is found to expand its distributional range dramatically when the **competing species** is experimentally removed. This is due to the phenomenon called '**competitive release**'. **Connell's** 'field experiments' showed that, on the rocky sea coasts of Scotland, the larger and competitively superior barnacle *Balanus* dominates the intertidal area, and excludes the smaller barnacle *Chthamalus* from that zone. When the dominant one is experimentally removed, the populations of the smaller ones increased. In general, herbivores and plants appear to be more adversely affected by competition than the carnivores (Ref: NCERT text book).

IV. Coexistence, rather than exclusion

Gause's principle of 'Competitive Exclusion' states that two closely related species competing for the same resources cannot co-exist indefinitely and the **competitively inferior one** will be eliminated in due course of time. This may be true if resources are limiting, but not otherwise.

More recent studies point out that species facing competition might evolve mechanisms that promote **co-existence** rather than competitive exclusion. One such mechanism is '**resource partitioning**'. If two species compete for the same resource, they could avoid competition by choosing, for instance, different times for feeding or different foraging (food collecting) patterns. **MacArthur** showed that five closely related species of warblers (a kind of birds) living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities.

8.3.4 Parasitism

Considering that the parasitic mode of life ensures free 'lodging' and 'meals', it is not surprising that parasitism has evolved in so many taxonomic groups from plants to higher vertebrates. Many parasites have evolved to be **host-specific** (they can parasitize only a specific species of host) in such a way that both host and the parasite tend to **co-evolve**; that is, if the host evolves special mechanisms for rejecting or resisting the parasite, the parasite has to evolve mechanisms to 'counteract' and 'neutralize' them, in order to continue successful parasitic relationship with the same host species. In order to lead successful parasitic life, parasites evolved special adaptations, such as:

- a. Loss of sense organs (which are not necessary for most parasites).
- b. Presence of adhesive organs such as suckers, hooks to cling on to the host's body parts.
- c. Loss of digestive system and presence of high reproductive capacity.
- d. The life cycles of parasites are often **complex**, involving one or two intermediate hosts or vectors to facilitate parasitisation of their primary hosts.

e.g.-1: The human liver fluke depends on two **intermediate** (secondary) **hosts** (a **snail** and a **fish**) to complete its life cycle.

e.g.-2: The malaria parasite needs a vector (mosquito) to spread to other hosts.

Majority of the parasites harm the host; they may reduce the survival, growth and reproduction of the host and reduce its population density. They might render the host more vulnerable to predation by making it physically weak.

The Types of Parasites

- I. Ectoparasites:** These parasites feed on host organism while remaining outside the body of the host. The most familiar examples of this group are the **lice** on humans and **ticks** on dogs. Many marine fish are infested with ectoparasitic copepods.

Plant examples: *Cuscuta*, a parasitic plant that is commonly found growing on hedge plants, has lost its chlorophyll and leaves in the course of evolution. It derives its nutrition from the host plant which it parasitises. (Ref: NCERT Text Book)

- II. Endoparasites:** Endoparasites are those that live inside the host's body at different sites (liver, kidney, lungs, red blood cells, etc.). The life cycles of endoparasites are more complex because of their extreme specialization. Their morphological and anatomical features are greatly simplified while emphasizing (stressing on) their reproductive potential.
- III. Brood parasitism:** Certain birds show a special type of parasitism, in which the 'parasitic bird' lays its eggs in the nest of its host and lets / allows the host to incubate them. During the course of evolution, the eggs of the parasitic bird have evolved to resemble the host's egg in size and colour to reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest. This is treated as a special type of parasitism by some

Try to follow the movements of the cuckoo (koel) and the crow in your neighborhood park during the breeding season (spring to summer) and watch brood parasitism in action (NCERT Text Book).

8.3.5 Commensalism

This is the interaction in which one species benefits and the other is neither harmed nor benefited. Barnacles growing on the back of a whale benefit while the whale derives no noticeable benefit.

The ‘cattle egret’ (a kind of bird) and ‘grazing cattle’ living in close association, is a good example of commensalism. The egrets always forage close to where the cattle are grazing because the cattle, as they move, stir up and drive out insects, from the vegetation thus helping the egrets to find and catch the insects. Another example of commensalism is the interaction between **sea anemone** that has tentacles with **stinging cells** and the ‘**clown fish**’ that lives among them. The fish gets protection from predators, as the predator stays away from the stinging tentacles of the anemone. The anemone does not derive any visible benefit by ‘hosting’ the clown fish.

Plant example :An orchid growing as an epiphyte on a mango branch, gets the benefit of exposure to light, while the mango tree does not derive any noticeable benefit (Ref: NCERT Text Book)

8.3.6 Mutualism

This type of interaction benefits both the interacting species.

The most common examples of mutualism are found in plant-animal relationships. Plants need the help of animals for **pollinating** their flowers and **dispersing** their seeds. Animals obviously have to be paid ‘**fees**’ for the services that plants derive from them. Plants offer rewards in the form of **pollen** and **nectar** for pollinators and juicy and nutritious **fruits** for seed dispersing animals.

Now you can see why plant-animal interactions often involve **co-evolution** of the mutualists, that is, the evolutions of the flower and its pollinator species are tightly linked with one another.



Fig. 8.8 Commensalism

Plant Examples: Lichens represent an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria. Similarly, the **mycorrhizae** are associations between fungi and the roots of higher plants. The fungi help the plant in the absorption of essential nutrients from the soil while the plant in turn provides the fungi with energy-yielding carbohydrates (Ref: NCERT Text Book).



Fig. 8.9 Mutual relationship between fig tree and wasp (a) Fig flower is pollinated by wasp; (b) Wasp laying eggs in a fig fruit

In many species of fig trees, there is a one-to-one relationship with the pollinator species of **wasp**. It means that a given fig species can be pollinated only by its '**partner**' **wasp** species and no other species. The female wasp uses the fruit not only as a site for oviposition (**egg-laying site**), but also uses the developing seeds within the fruit for nourishing its larvae. The wasp pollinates the flowers of the fig plant while searching for suitable egg-laying sites.

In return for the favour of pollination the fig offers the wasp some of its developing seeds, as food for the developing wasp larvae.



Fig. 8.10 Showing bee a pollinator on orchid flower

Orchids show a high degree of diversity of floral patterns. Many of them have evolved to attract the right pollinator insect (bees and bumblebees) and ensure pollination by it. Not all orchids offer rewards. The Mediterranean orchid *Ophrys* employs '**sexual deceit**' to get pollination done by a specific species of bee. One petal of its flower resembles the female of the bee in size, colour and markings. The male bee is attracted to the 'petal' as it presumes it to be the female, '**pseudocopulates**' with the flower, and during that process, it is dusted with pollen from the flower. When the same bee '**pseudocopulates**' with another flower, it transfers the pollen to it. Here you can see how **co-evolution** operates in nature. If the female bee's colour patterns change even slightly, for any reason in the course of evolution, pollination success will be reduced, unless the orchid flower **co-evolves** and alters the resemblance of its petal to the changed female bee.

8.4. ECOSYSTEMS AND THEIR COMPONENTS

- 8.4.1 The Natural Ecosystems
- 8.4.2 Aquatic Ecosystems
- 8.4.3 Lake Ecosystem
- 8.4.4 Terrestrial Ecosystems
- 8.4.5 Artificial Ecosystems

An 'ecosystem' is a **functional unit of nature**, where living organisms interact among themselves and also with the surrounding physical environment. Ecosystem varies greatly in size from a small pond to a large forest or a sea. Many ecologists regard the entire **biosphere** as a 'global ecosystem', as a composite of all local ecosystems on Earth. Since this system is too big and complex to be studied at one time, it is convenient to divide it into two basic categories, namely natural and artificial. The natural ecosystems include **aquatic ecosystems** of water and **terrestrial ecosystems** of the land. Both types of natural and artificial ecosystems have several subdivisions.

8.4.1 The Natural Ecosystems

These are naturally occurring ecosystems and there is no role of humans in the formation of such types of ecosystems. These are categorized mainly into two types - aquatic and terrestrial ecosystems.

8.4.2 Aquatic Ecosystems

Based on the **salinity** of water, three types of aquatic ecosystems are identified- marine, freshwater, and estuarine.

- i. **The Marine Ecosystem:** It is the largest of all the aquatic ecosystems. It is the most stable ecosystem
- ii. **Estuarine Ecosystem:** Estuary is the zone where river joins the sea. Sea water ascends up into the river twice a day (effect of high tides and low tides). The salinity of water in an estuary also depends on the seasons. During the rainy season out flow of river water makes the estuary less saline and the opposite occurs during the summer. Estuarine organisms are capable of withstanding the '**fluctuations**' in salinity.
- iii. **The Freshwater Ecosystem:** The freshwater ecosystem is the smallest aquatic ecosystem. It includes rivers, lakes, ponds, etc., It is divided into two groups- the **lentic** and **lotic**. The still water bodies like ponds, lakes, reservoirs, etc., fall under the category of **lentic ecosystems**, whereas, streams, rivers and flowing water bodies are called **lotic ecosystems**. The communities of the above two types are called lentic and lotic communities respectively. The study of freshwater ecosystem is called as **limnology**.

8.4.3 Lake Ecosystem

To understand the fundamentals of an aquatic ecosystem, let us take a 'lake' as an example. This is fairly a self-sustainable unit and rather a simple example that explains even the complex interactions that exist in an aquatic ecosystem.

Lakes are large inland water bodies containing standing/still water (Recall: Lentic community). They are deeper than ponds (pond is not an ideal example as it is very shallow). Most lakes contain water throughout the year. In deep lakes, light cannot penetrate more than 200 meters, in depth. They are vertically stratified in relation to light intensity, temperature, pressure, etc., Deep water lakes contain three distinct zones namely, i) littoral zone, ii) limnetic zone, and iii) profundal zone.

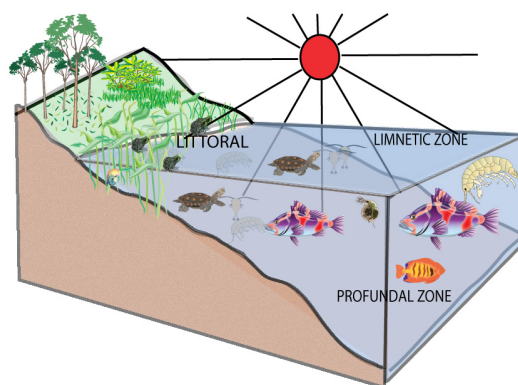


Fig. 8.11 Lake Ecosystem

- ❖ **Littoral zone:** It is the shallow part of the lake closer to the shore. Light penetrates up to the bottom. It is 'euphotic' (having good light), has rich vegetation and higher rate of photosynthesis, hence rich in oxygen.
- ❖ **Limnetic zone:** It is the open water zone away from the shore. It extends up to the **effective light penetration level**, vertically. The imaginary line that separates the limnetic zone from the profundal zone is known as **zone of compensation/ compensation point/ light compensation level**. It is the zone of effective light penetration. Here the rate of photosynthesis is equal to the rate of respiration. Limnetic zone has no contact with the bottom of the lake.
- ❖ **Profundal zone:** It is the deep water area present below the limnetic zone and beyond the depth of effective light penetration. Light is absent. Photosynthetic organisms are absent and so the water is poor in oxygen content. It includes mostly the anaerobic organisms which feed on detritus.

The organisms living in lentic habitat are classified into **pedonic** forms, which live at the bottom of the lake and those living in the open waters of lakes, away from the shore vegetation are known as **limnetic forms**.

Biota (animal and plant life of a particular region) of the littoral zone

Littoral zone is rich with pedonic flora (especially up to the depth of the effective light penetration). At the shore proper '**emergent vegetation**' is abundant with firmly fixed roots in the bottom of the lake and shoots and leaves are exposed above the level of water. These are **amphibious plants**. Certain emergent rooted plants of littoral zone are the cattails (*Typha*), bulrushes (*Scirpus*), arrowheads (*Sagittaria*). Slightly deeper are the rooted plants with floating leaves, such as the water lilies (*Nymphaea*), *Nelumbo*, *Trapa*, etc. Still deeper are the submerged plants such as *Hydrilla*, *Chara*, *Potamogeton*, etc., The free floating vegetation includes *Pistia*, *Wolffia*, *Lemna* (duckweed), *Azolla*, *Eichhornia*, etc.

The **phytoplankton** of the littoral zone composed of diatoms (*Coscinodiscus*, *Nitzschia*, etc.), green algae (*Volvox*, *Spirogyra*, etc.), euglenoids (*Euglena*, *Phacus*, etc.), and dinoflagellates (*Gymnodinium*, *Cystodinium*, etc.).

Animals, the **consumers** of the littoral zone, are abundant in this zone of the lake. These are categorized into zooplankton, neuston, nekton, periphyton, and benthos. The **zooplankton** of the littoral zone consists of 'water fleas' such as *Daphnia*, rotifers and ostracods.

The animals living at the *air-water interface* constitute the '**neuston**'. They are of two types, the epineuston and hyponeuston. Water striders (*Gerris*), beetles, water bugs (*Dineutes*) form the **epineuston/supraneuston** and the **hyponeuston/infraneuston** includes the 'larvae of mosquitoes'.

The animals such as fishes, amphibians, water snakes, terrapins, insects like 'water scorpion' (*Ranatra*), 'back swimmer' (*Notonecta*), 'diving beetles' (*Dytiscus*), capable of swimming constitute the **nekton**.

The animals that are attached to / creeping on the aquatic plants, such as the 'water snails', 'nymphs of insects', 'bryozoans', 'turbellarians', 'hydras', etc., constitute the '**periphyton**'.

The animals that rest on or move on the bottom of the lake constitute the '**benthos**', e.g., red annelids, chironomid larvae, cray fishes, some isopods, amphipods, clams, etc.

Biota of the limnetic zone

Limnetic zone is the largest zone of a lake. It is the region of rapid variations of the level of the water, temperature, oxygen availability, etc., from time to time. The limnetic zone has autotrophs (photosynthetic plants) in abundance. The chief autotrophs of this region are the phytoplankton such as the **euglenoids, diatoms, cyanobacteria, dinoflagellates** and **green algae**. The consumers of the limnetic zone are the zooplanktonic organisms such as the copepods. Fishes, frogs, water snakes, etc., form the limnetic **nekton**.

Biota of the profundal zone

It includes the organisms such as decomposers (bacteria), *chironomid* larvae, *Chaoborus* (**phantom larva**), red annelids, clams, etc., that are capable of living in low oxygen levels. The decomposers of this zone decompose the dead plants and animals and release nutrients which are used by the biotic communities of both littoral and limnetic zones.

The lake ecosystem performs all the functions of any ecosystem and of the biosphere as a whole, i.e., conversion of inorganic substances into organic material, with the help of the radiant solar energy by the autotrophs; consumption of the autotrophs by the heterotrophs; decomposition and mineralization of the dead matter to release them back for reuse by the autotrophs (**recycling of minerals**).

8.4.4 The Terrestrial Ecosystems

The ecosystems of land are known as terrestrial ecosystems. Some examples of terrestrial ecosystems are the forest, grassland and desert.

- i. The Forest Ecosystems:** The two important types of forests seen in India are i) tropical rain forest and ii) tropical deciduous forests.
- ii. The Grassland Ecosystems:** These are present in the Himalayan region in India. They occupy large areas of sandy and saline soils in Western Rajasthan.
- iii. Desert Ecosystems:** The areas having less than 25cm rainfall per year are called deserts. They have characteristic flora and fauna. The deserts can be divided into two types – hot type and cold type deserts. Thar Desert in Rajasthan is the example for hot type of desert. Cold type desert is seen in Ladakh.

8.4.5 Artificial Ecosystems

These are man-made ecosystems such as agricultural or agro-ecosystems. They include cropland ecosystems, aquaculture ponds and aquaria.

8.5 FOOD CHAINS, FOOD WEB, PRODUCTIVITY AND ENERGY FLOW

- 8.5.1 Biotic factors
- 8.5.2 Food chains
- 8.5.3 Energy Flow
- 8.5.4 Productivity
- 8.5.5 Ecological pyramids

An ecosystem consists of structural and functional components. The structural components are **abiotic** and **biotic** factors. You have studied various abiotic components like light, temperature, water, pressure etc., of ecosystem in the previous lessons. The biotic components of an ecosystem are composed of producers, consumers, and decomposers. The functional components of ecosystem are **cycling of nutrients** and the **flow of energy**. Let us look at these components in a more integrated manner and see how the flow of energy takes place within these components of the ecosystem.

8.5.1 Biotic Factors

1. Producers

The green plants in the ecosystem-terminology are called **producers**. They synthesize the food by using solar energy (photo autotrophs). Certain bacteria such as Iron and Sulphur bacteria are **chemoautotrophs**. They fix carbon dioxide by using the energy obtained by the breaking down of certain chemical substances. In a terrestrial ecosystem, the major producers are the herbaceous and woody plants. Likewise, primary producers in an aquatic ecosystem include various species like microscopic phytoplanktonic organisms, algae and some higher plants. The producers form the first trophic level in most food chains. They constitute the main source of food for the organisms of the second/ next higher trophic level, the 'consumers'.

2. Consumers

All animals depend on plants (directly or indirectly) for their food needs. They are hence called consumers and also **heterotrophs**. If they feed on the producers - (the plants) they are called **primary consumers**. If the animals eat other animals, which eat plants or plant products, they are called **secondary consumers**. Likewise, there may be **tertiary consumers** too. Obviously the primary consumers are the herbivores. Some common herbivores are insects, birds and mammals in the terrestrial ecosystem and arthropods and molluscs in the aquatic ecosystems.

3. Decomposers

You may have heard of the earthworm being referred to as the farmer's 'friend'. This is so because they help in the breakdown of complex organic matter as well as in loosening of the soil (making the soil porous). Similarly, decomposers break down complex organic matter into simple inorganic substances like carbon dioxide, water and nutrients and the process is called **decomposition**. Plant parts such as leaves, bark, flowers and dead remains of animals, including their faecal matter, constitute the **detritus**. Detritus is the '**raw material**' for '**decomposition**'.

The important steps in the process of decomposition are **fragmentation**, **leaching**, **catabolism**, **humification** and **mineralization**.

Detritivores (e.g. earthworm) break down detritus (litter) into smaller particles and this process is called **fragmentation**. By the process of **leaching**, water soluble inorganic nutrients go down into the soil and get precipitated as unavailable salts. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called **catabolism**.

It is important to note that all the above steps in decomposition operate simultaneously on the detritus. Humification and mineralization occur during decomposition in the soil. Humification leads to accumulation of a dark coloured amorphous substance called **humus** that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate. Being colloidal in nature it serves as a **reservoir of nutrients**. The humus is further degraded by some microbes and release of inorganic nutrients occurs by the process known as **mineralization**. (Ref: NCERT Text Book pages 243-244)

Decomposition

Decomposition is largely an oxygen-requiring process. The rate of decomposition is controlled by chemical composition of the detritus and *climatic factors*. In a particular climatic condition, decomposition rate is slower if detritus is rich in **lignin**.

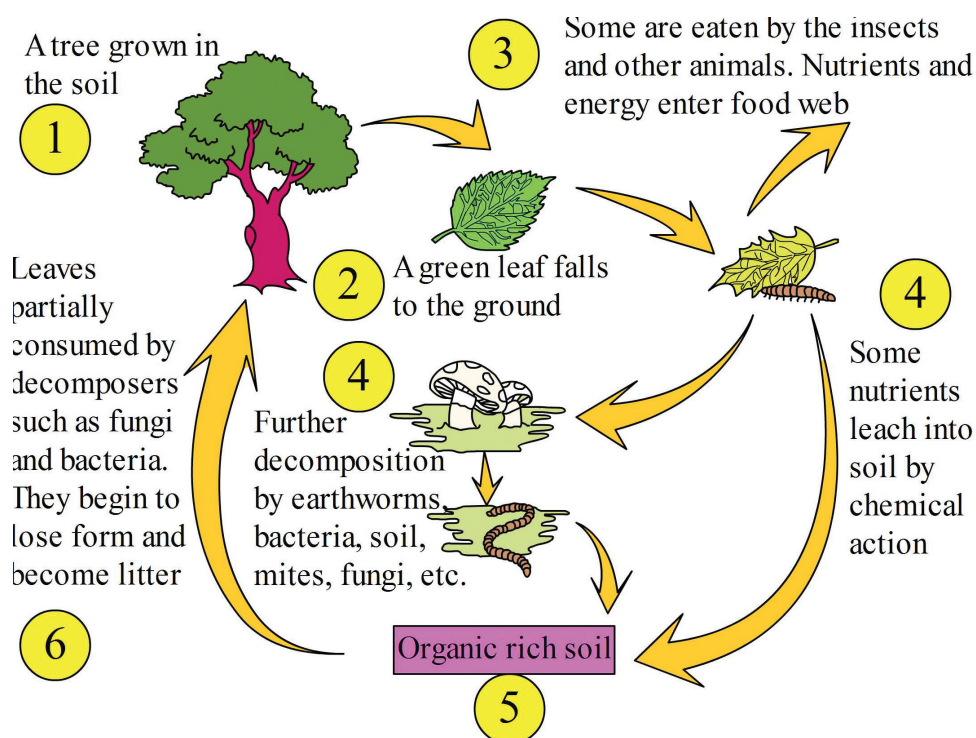
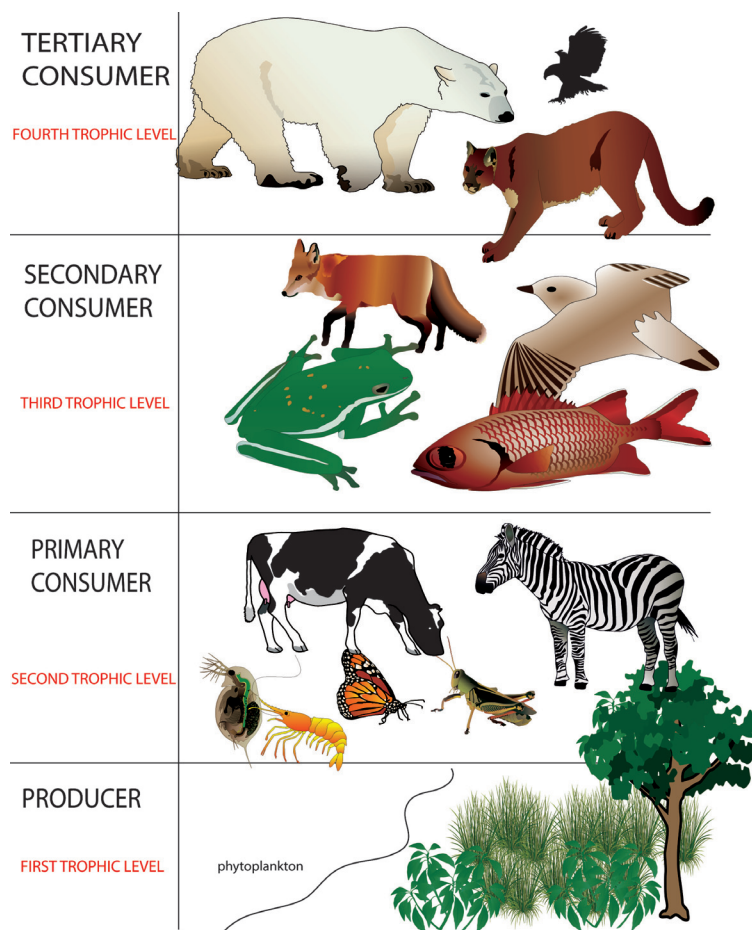


Fig. 8.12 Diagrammatic representation of decomposition cycle in a terrestrial ecosystem (NCERT)

and **chitin**, and quicker, if detritus is rich in **nitrogen** and water-soluble substances like **sugars**. Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes. Warm and moist environment favours decomposition whereas **low temperature** and **anaerobic environment** 'inhibit' the decomposition resulting in build up of organic materials. As most of the decomposers are very small microscopic forms, they are also called 'micro-consumers'.

8.5.2 The Food Chains

Energy flows into biological systems (ecosystems) from the Sun. The biological systems of environment include several food levels called **trophic levels**. A trophic level is composed of those organisms which have the same source of energy and having the same number of steps (energy storing and transfer steps) away from the sun. Thus a plant's trophic level is **one**, while that of a herbivore - **two**, and that of the first level carnivore - **three**. The second and third levels of the carnivores occupy **fourth** and **fifth** trophic levels respectively.



A given organism may occupy more than one trophic level simultaneously because of its food habits. One must remember that the trophic level represents a functional level. A given species may occupy more than one trophic level in the same ecosystem at the same time; for example, a sparrow is a primary consumer when it eats seeds, fruits, and a secondary consumer when it eats insects and worms (omnivore).

The food energy passes from one trophic level to another trophic level mostly from the lower to higher trophic levels. When the path of food energy is 'linear', the components resemble the 'links' of a chain, and it is called '**food chain**'. Generally a food chain ends with decomposers. The three major types of food chains in an ecosystem are Grazing Food Chain, Parasitic Food Chain and Detritus Food Chain.

Fig. 8.13 Trophic levels in an ecosystem

I. Grazing Food Chain

It is also known as predatory food chain. It begins with the green plants (producers) and the second, third and fourth trophic levels are occupied by the herbivores, primary carnivores and secondary carnivores respectively. In some food chains there is yet another trophic level – the climax carnivores. The number of trophic levels in food chains varies from 3 to 5 generally. Some examples for grazing food chain (**GFC**) are given below.

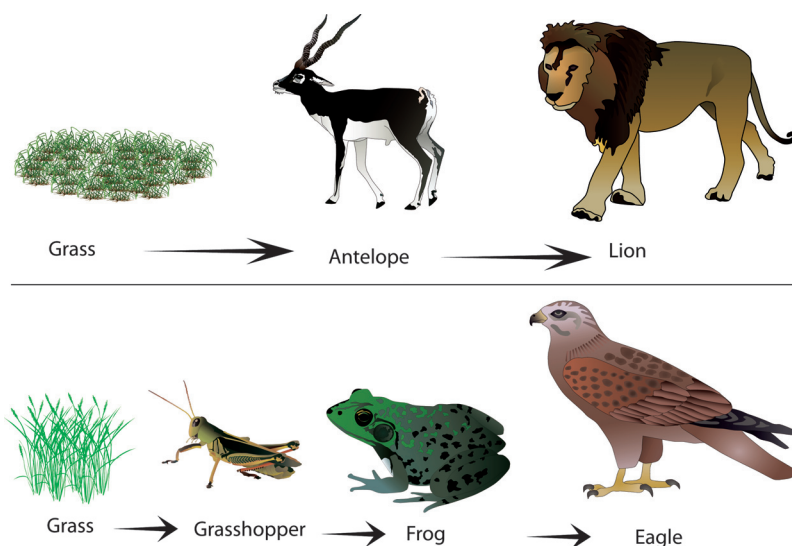


Fig. 8.14 Grazing food chain

I trophic level	II trophic level	III trophic level	IV trophic level	V trophic level
Rosebush	→ Aphids	→ Spiders	→ Small birds	→ Hawk
Grass	→ Grasshopper	→ Frog	→ Snake	→ Hawk
Plants	→ Caterpillar	→ Lizard	→ Snake	
Phytoplankton	→ Zooplankton	→ Fish	→ Bird	
Grass	→ Goat	→ Man		

II. Parasitic food chain

Some authors included the 'Parasitic Food Chains' as a part of the GFC. As in the case of GFCs, it also begins with the producers, the plants (directly or indirectly). However, the food energy passes from large organisms to small organisms in the parasitic chains. For instance, a tree which occupies the 1st trophic level provides shelter and food for many birds. These birds host many **ecto-parasites** and **endo-parasites**. Thus, unlike in the predator food chain, the path of the flow of energy includes fewer, large sized organisms in the lower trophic levels, and numerous, small sized organisms in the successive higher trophic levels.

III. Detritus Food Chain

The detritus food chain (**DFC**) begins with dead organic matter (such as leaf litter, bodies of dead organisms and faeces of animals). It is made up of **decomposers** which are **heterotrophic organisms**, mainly the 'fungi' and 'bacteria'. They meet their energy

and nutrient requirements by degrading dead organic matter or detritus. These are also known as **saprotrophs** (sapro: to decompose).

Decomposers secrete digestive enzymes that breakdown dead and waste materials (such as faeces) into simple absorbable substances. Some examples of detritus food chains are:

1. Detritus (formed from leaf litter)-Earthworms -Frogs -Snakes
2. Dead animals -Flies and maggots -Frogs -Snakes

In an aquatic ecosystem, GFC is the major ‘conduit’ for the energy flow. As against this, in a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFCs. Detritus food chain may be connected with the grazing food chain at some levels. Some of the organisms of DFC may form the prey of the **GFC** animals. For example, in the detritus food chain given above, the earthworms of the **DFC** may become the food of the birds of the **GFC**. It is to be understood that food chains are not ‘isolated’ always.

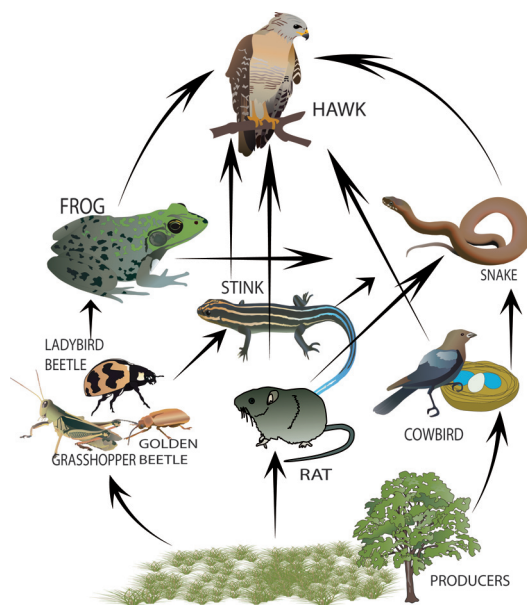


Fig. 8.15 A simple food web

8.5.2.1 Food Web

In a simple GFC the food energy flows through in **linear trophic levels**. However, in an ecosystem the feeding relationships are never that simple. The feeding interrelationships become more complex because of the omnivores present in the ecosystem. For instance man, birds and some other animals are ‘**omnivores**’, by their feeding habits. In such cases it is difficult to place an animal in a simple / single food chain as they have many interconnections with the animals of other food chains. These natural interconnections of food chains form a ‘net-work’ called ‘**food web**’.

8.5.3 Energy Flow

Except for the deep sea **hydro-thermal ecosystem**⁷, sun is the only source of energy for all ecosystems on Earth. Of the incident solar radiation less than 50 per cent of it is **photosynthetically active radiation** (PAR). We know that plants and photosynthetic bacteria (autotrophs), fix Sun’s radiant energy to synthesise food from simple inorganic materials. Plants capture only 2-10 percent of the PAR and this small amount of energy sustains the entire living world. So, it is very important to know how the solar energy captured by plants flows through different organisms of an ecosystem. All heterotrophs are dependent on the producers for their

⁷ Hydrothermal systems in the deep ocean, do not have light, and chemical energy supports life (chemoautotrophs), instead of sunlight.

food, either directly or indirectly. The law of conservation of energy is the **first law of thermodynamics**. It states that energy may transform from one form into another form, but it is neither created nor destroyed. The energy that reaches earth is balanced by the energy that leaves the surface of the earth as invisible heat radiation.

The energy transfers in an ecosystem are essential for sustaining life. Without energy transfers there could be no life and ecosystems. Living beings are the natural proliferations that depend on the continuous inflow of concentrated energy.

Further, ecosystems are not exempted from the Second Law of thermodynamics. It states that no process involving energy transformation will spontaneously occur unless there is degradation of energy. As per the second law of thermodynamics - the energy degraded is in the form of unavailable heat energy, and constitutes the **entropy** (energy lost or not available for work in a system). The organisms need a constant supply of energy to synthesize the molecules they require. The transfer of energy through a food chain is known as **energy flow**. A constant input of mostly solar energy is the basic requirement for any ecosystem to function. The important point to note is that the amount of energy available decreases at successive trophic levels. When an organism dies, it is converted to detritus or dead biomass that serves as a source of energy for the decomposers. Organisms at each trophic level depend on those at the lower trophic level, for their energy demands.

Each trophic level has a certain mass of living material at a particular time, and it is called the **standing crop**. The standing crop is measured as the **mass** of living organisms (**biomass**) or the number of organisms per unit area. The biomass of a species is expressed in terms of fresh or dry weight (dry weight is more accurate – because water contains no usable energy).

The 10 percent Law

The 10 percent law for the transfer of energy from one trophic level to the next was introduced by Lindeman (the Founder of the modern Ecosystem Ecology). According to this law, during the transfer of energy from one trophic level to the next, only about 10 percent of the energy is stored/ converted as body mass / biomass. The remaining is lost during the transfer or broken down in catabolic activities (Respiration). Lindeman's rule of trophic efficiency/ Gross ecological efficiency is one of the earliest and most widely used measures of ecological efficiency. For example, if the NPP (Net primary production) in a plant is 100 kJ, the organic substance converted into **body mass** of the herbivore which feeds on it is 10 kJ only. Similarly the **body mass** of the carnivore-I is 1 kJ only.

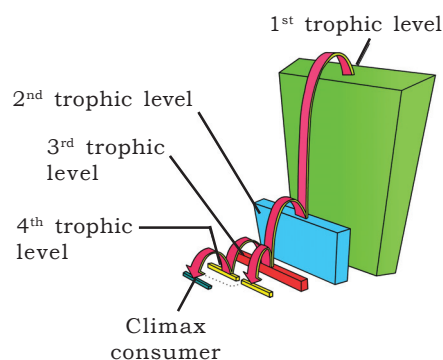


Fig. 8.16 Energy flow in an ecosystem

8.5.4 Productivity

The **rate of production** of biomass is called **productivity**. It can be divided into **primary** and **secondary** productivities.

- I. **Primary productivity** is defined as the amount of biomass or organic matter produced per unit area over a period of time by plants, during photosynthesis. It can be divided into gross primary productivity (GPP) and net primary productivity (NPP).
 - a) **Gross primary productivity** of an ecosystem is the rate of production of organic matter during photosynthesis. A considerable amount of GPP is utilized by plants for their catabolic process (respiration).
 - b) **Net primary productivity** Gross primary productivity minus respiratory loss (R), is the net primary productivity (NPP). On average about 20-25 percent of GPP is used for the catabolic (respiratory) activity.

GPP - R = NPP

The net primary productivity is the biomass available for the consumption of the heterotrophs (herbivores and decomposers).

- II. **Secondary productivity** is defined as the rate of formation of new organic matter by consumers.

8.5.5 Ecological Pyramids (Eltonian pyramids)

You must be familiar with the shape of a pyramid. The base of a pyramid is broad and it narrows down towards the apex. The trophic relationship is expressed in terms of numbers; biomass or energy arranged one on the top of its lower trophic level, resulting in a pyramidal shape. It is a graphical representation of the trophic structure and function of an ecosystem. The base of each pyramid represents the **producers** or the first trophic level, while the apex represents the tertiary or top level / top order consumer. The three types of ecological pyramids that are usually studied are (a) **pyramid of number**; (b) **pyramid of biomass** and (c) **pyramid of energy**. These pyramids were first represented by **Elton**, hence the name **ELTONIAN pyramids/ Ecological pyramids**.

Any calculations of energy content, biomass, or numbers has to include all organisms at that trophic level. No generalizations we make will be true if we take only a few individuals of any trophic level into account. In most ecosystems, all the pyramids- of numbers, energy and biomass are **upright** i.e., producers are more in number and biomass than the herbivores, and herbivores are more in number and biomass than the carnivores. Also energy (available) at a lower trophic level is always more than that at a higher level.

There are exceptions to this generalization. In the case of a parasitic food chain, the pyramid of numbers is **inverted**. A large tree (single producer) may support many

herbivores like squirrels, and fruit eating birds. On these herbivores many ectoparasites such as ticks, mites and lice (secondary consumers) may live. These secondary consumers may support many more top level consumers and also the hyperparasites. Thus in each trophic level from the bottom to the top, the numbers of organisms increase, and form an 'inverted pyramid' of **numbers**.

The pyramid of **biomass** in sea is also generally inverted because the biomass of fishes far exceeds that of phytoplankton (Ref: NCERT Text Book).

Pyramid of **energy** is always **upright**, and can never be inverted, because when energy flows from a particular trophic level to the next higher trophic level, some energy is always lost as heat (thus at every step). Each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time or annually per unit area.

Limitations of Ecological Pyramids

However, there are certain limitations of ecological pyramids, such as –

- it does not take into account the same species belonging to two or more trophic levels,
- It assumes a simple food chain, something that almost never exists in nature,
- it does not accommodate a food web, iv) moreover, saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

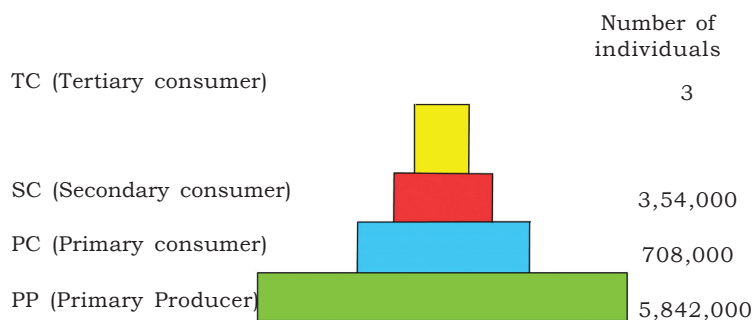


Fig. 8.17 (a) Pyramid of Numbers in a grassland ecosystem. NOTE: Only 3 top-carnivores are supported in an ecosystem based on production of nearly 6 million plants. Also NOTE that the pyramid of numbers of a parasitic food chain is inverted.

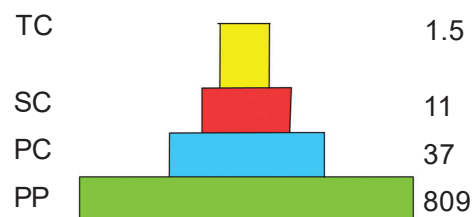


Fig. 8.17 (b) Pyramid of biomass shows a sharp decrease in biomass at the higher trophic levels. NOTE: Pyramid of biomass is upright normally.

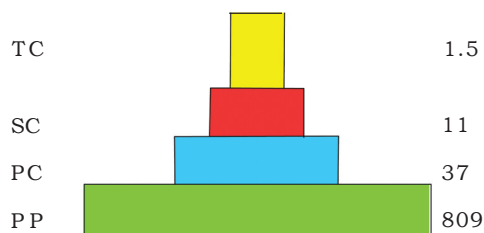


Fig. 8.17 (c) An ideal pyramid of energy which is an always upright pyramid. NOTE: primary producers convert only 0.5% of the energy absorbed from the sunlight available to them are into GPP

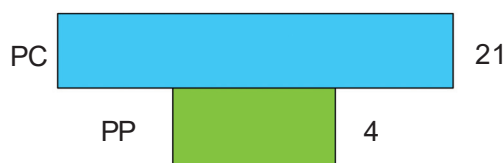


Fig. 8.17 (d) Inverted pyramid of biomass-small standing crop of phytoplankton supports large standing crop of zooplankton (at certain times) NOTE: Algal blooms are followed by their death due to short life span depleting their numbers (temporarily).

8.6 NUTRIENT CYCLES

8.6.1 Carbon Cycle

8.6.2 Nitrogen Cycle

8.6.3 Phosphorous Cycle

Organisms need a constant supply of nutrients to grow, reproduce and regulate various body functions. The amount of nutrients, such as carbon, nitrogen, phosphorus, calcium, etc., present in the soil at any given time, is referred to as the **standing state**. It varies in different kinds of ecosystems and also on a seasonal basis.

Nutrients are never lost from the ecosystems. They are recycled again and again, indefinitely. The movement of nutrient elements through the various components of an ecosystem is called '*nutrient cycling*'. Such cycles are called biogeochemical cycles (bio: living organism, geo: rocks, air, water).

Nutrient cycles are of two types:

- (a) gaseous and
- (b) sedimentary.

In a gaseous cycle, elements move through the atmosphere. Main reservoirs are atmosphere and oceans (via evaporation). e.g. carbon cycle and nitrogen cycle.

In a sedimentary cycle, elements move from the earth's crust to water and to sediment. Main reservoirs are the soil and sedimentary rocks. e.g. phosphorous cycle and sulphur cycle.

8.6.1 Carbon Cycle

The element carbon constitutes 49 percent of the dry weight of organisms and is next only to water. Among the total carbon quantity present on the Earth, 71 percent is found dissolved in oceans. This 'oceanic reservoir' regulates the amount of carbon dioxide in the atmosphere. It will be interesting to know that the atmosphere contains only about 1 percent of the total global carbon.

'**Fossil fuels**' also represent reservoirs of carbon. Carbon cycling occurs through atmosphere, ocean and through living and dead organisms. carbon is fixed in the biosphere through photosynthesis. A considerable amount of carbon returns to the atmosphere as CO_2 through respiratory activities of the producers and consumers. Decomposers also contribute substantially to CO_2 pool by their processing of waste materials and dead organic matter of land or oceans. Some amount of the fixed carbon is lost to sediments and removed from circulation. Burning of wood, forest fires combustion of organic matter, fossil fuels, volcanic activity etc., are additional sources for releasing CO_2 into the atmosphere.

Human activities have significantly influenced the carbon cycle. Rapid deforestation and massive burning of fossil fuel for energy and transport have significantly increased the rate of release of carbon dioxide into the atmosphere. (see greenhouse effect at 8.8.4)

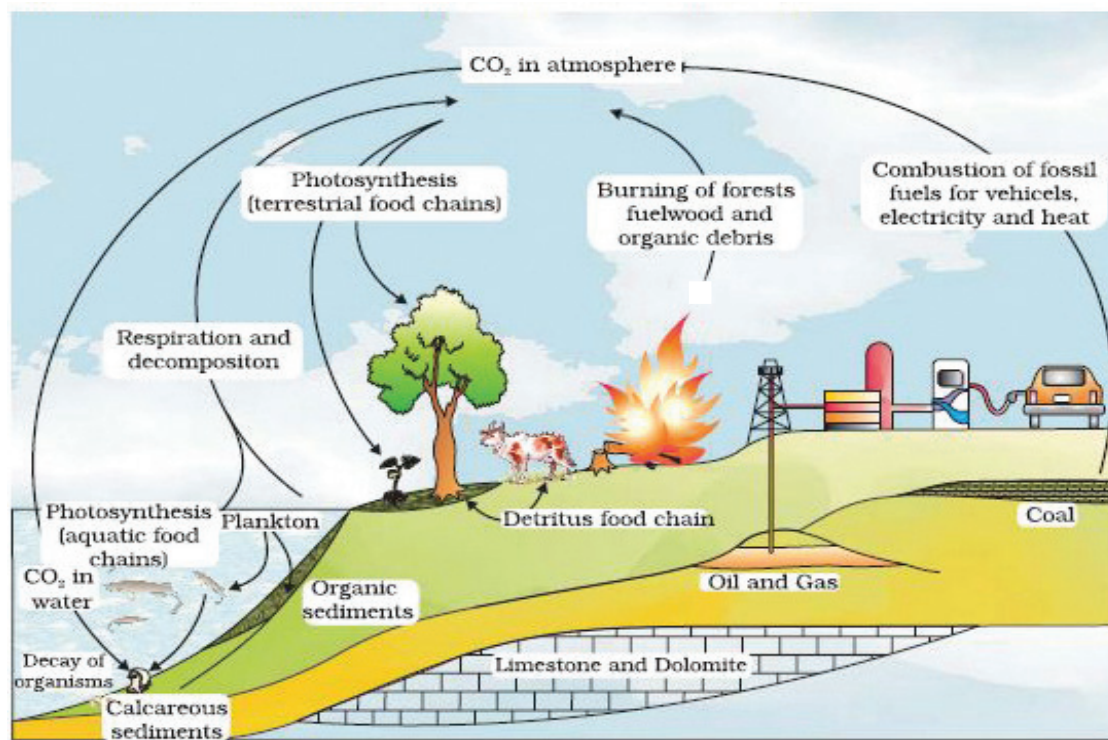


Fig. 8.18 Simplified model of carbon cycle in the biosphere

8.6.2 Nitrogen Cycle

Apart from carbon, hydrogen and oxygen, nitrogen is the most prevalent element in living organisms. Nitrogen is a constituent of amino acids, proteins, hormones, chlorophylls and many of the vitamins. Plants compete with microbes for the limited nitrogen that is available in the soil. Thus nitrogen is a '**limiting nutrient**' for both natural and agricultural ecosystems.

Nitrogen exists in molecular state. The process of conversion of nitrogen (N₂) into nitrites and nitrates is called **nitrogen fixation**. In nature lightening and ultra violet radiation provide enough energy to convert nitrogen to nitrogen oxides NO, NO₂,

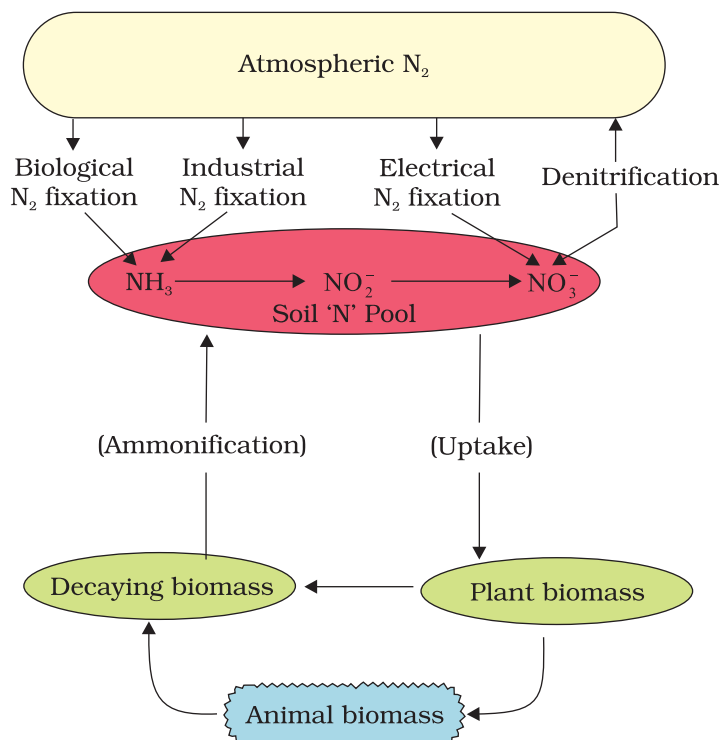
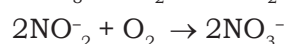


Fig. 8.19 The nitrogen cycle showing relationship between the three main nitrogen pools – atmosphere, soil and biomass

N_2O . Industrial combustions, forest fires, auto-mobile exhausts and power generating stations are also sources for atmospheric nitrogen oxides. Decomposition of organic nitrogen of dead plants and animals into ammonia is called 'ammonification'. Most of the ammonia is converted into nitrites and nitrates by soil bacteria by the following reaction.



Ammonia is first oxidized to nitrite by bacteria such as *Nitrosomonas* and *Nitrosococcus*. Nitrites are further oxidized to 'nitrates' with the help of bacteria such as '*Nitrobacter*'. These steps constitute 'nitrification'. These nitrifying bacteria are *chemoautotrophs*.

The nitrate thus formed is absorbed by plants and is transported to the leaves. In leaves it is reduced to ammonia that finally forms the 'amine' group of amino acids. Nitrates present in the soil are also reduced to nitrogen by the process of 'denitrification'. Denitrification is carried out by bacteria such as *Pseudomonas* and *Thiobacillus*.

8.6.3 Phosphorus Cycle

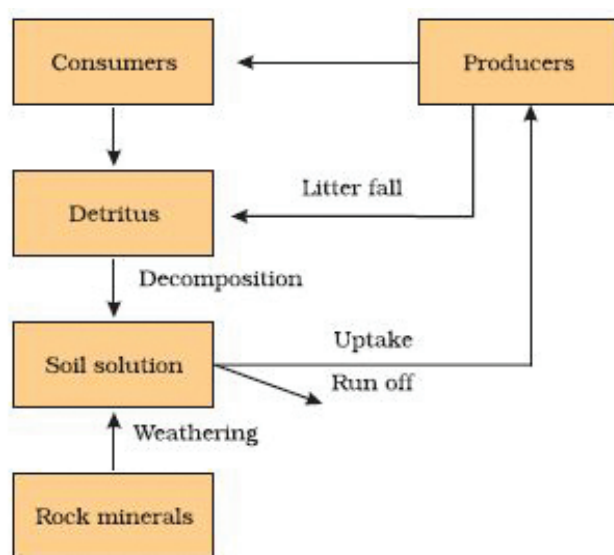


Fig. 8.20 A simplified model of phosphorus cycling in a terrestrial ecosystem

Phosphorus is a major constituent of biological membranes, nucleic acids and cellular energy transfer systems. Many animals also need large quantities of this element to make **shells**, **bones** and **teeth**. The natural reservoir of phosphorus is rock, which contains phosphorus in the form of 'phosphates'. When rocks are '**weathered**', minute amounts of these phosphates dissolve in soil solution and are absorbed by the roots of plants. Herbivores and other animals obtain this element from plants. The waste products and the dead organisms are decomposed by 'phosphate-solubilising bacteria' releasing phosphorus. Unlike carbon cycle, there is no respiratory release of phosphorus into the atmosphere.

8.7 POPULATION

8.7.1 Population attributes

8.7.2 Population growth

8.7.3 Life history variation

In nature, animals of any species which live as isolated single individuals are very rare. Majority of them live in groups, in a well defined geographical area. They share or compete for similar resources and interbreed to produce fertile offspring. Such a group of individuals is called a population. Though the term interbreeding implies sexual reproduction, generally a group of individuals resulting from even asexual reproduction is also considered a '**population**' in ecological terms. e.g. all the frogs living in a lake, rats in an abandoned dwelling, teakwood trees in a forest, bacteria on a *culture plate*, lotus plants in a pond and Paramecia in a pond.

It is a well known fact that an individual organism is the one that has to adjust to a changed environment. But it is also important to know that Natural Selection operates at the level of population in evolving the desired traits. Population ecology is, therefore, an important area of ecology because it links ecology to population genetics and evolution.

8.7.1 Population Attributes

A population has certain attributes that an individual organism does not exhibit.

- 1. Natality (Birth rate) and Mortality (Death rate):** An individual may have birth and death, but a population has **birth rate (natality)** and **death rate (mortality)**. Natality is the number of new individuals produced in a unit time, per unit population. Mortality is the number of deaths that occurred in a unit time, per unit population. In a population these rates refer to *per capita* (per individual) births and deaths, respectively. The rate refers to change in numbers (increase or decrease) with respect to members of the population in unit time. Here is an example. If, in a pond there were 20 frogs last year and through reproduction 8 new frogs are added, taking the current population to 28, we calculate the birth rate as $8/20 = 0.4$ offspring per frog per year. If 4 individuals in a laboratory population of 40 fruit-flies died during a specified time interval, say a week, the death rate in the population during that period is $4/40 = 0.1$ individuals, per fruit-fly per week.
- 2. Immigration and Emigration:** **Immigration** is the number of individuals of the same species that have come into the habitat/population from elsewhere/another population during the time period under consideration. Immigration contributes to an increase in the population size.
Emigration is the number of individuals of population which left the habitat/population and gone elsewhere during the time period under consideration. Due to emigration the size of the population decreases.

3. **Sex ratio:** Another characteristic that influences the growth of a population is the *sex ratio*. An individual is either a male or a female, but a population has a sex ratio (e.g., 60 percent of the population are males and 40 percent population are females).
4. **Age distribution:** A population at any given time is composed of individuals of different ages. If the age distribution (percent individuals of a given age or age group) is plotted for the population, the resulting structure is called an '**age pyramid**'. For human population, the age pyramids generally show age distribution of males and females in a combined diagram. The shape of the pyramids reflect the 'status' of the population whether it is growing, stable or declining.

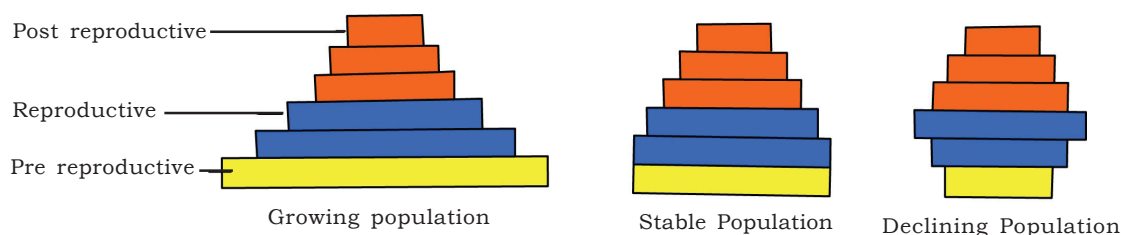


Fig. 8.21 Age pyramids

5. **Population size or Population density:** Population size, more technically called population density (designated as N) is the number of individuals per unit area (on land) or unit volume (in water).

Population density need not necessarily be measured in numbers only. Generally the most appropriate measure of population density is the total number. But in some cases it is either meaningless or difficult to determine. For example: Imagine there are 200 *Parthenium* plants and only a single huge banyan tree covering a large part in a particular area. If a statement is given that the population density of banyan is low relative to that of *Parthenium*, it amounts to underestimating the enormous role of the Banyan tree in that community. In such cases, the percent cover or biomass is a more meaningful measure of the population size. Total number is again not an easily adoptable measure if the population is huge and counting is impossible or very time-consuming. Sometimes, for certain ecological investigations, there is no need to know the absolute population densities; relative densities serve the purpose equally well. For instance, counting the number of fish caught per trap is good enough to measure its total population density in the lake. We mostly estimate population sizes indirectly, without actually counting them or seeing them. The tiger census in our National Parks and Tiger Reserves is often based on 'pug marks' and 'fecal pellets'. The size of the population provides a lot of information on the status of the species in the habitat. In the investigation of certain ecological processes in a population, change in the population size can be taken as an important parameter. e.g., the outcome of inter-specific competition, the impact of a predator, the effect of a pesticide application, etc., In nature, population size could be as low as <10 (Siberian

cranes at Bharatpur wetlands (Keoladeoghana National Park) in any year) or go into millions (*Chlamydomonas* in a pond).

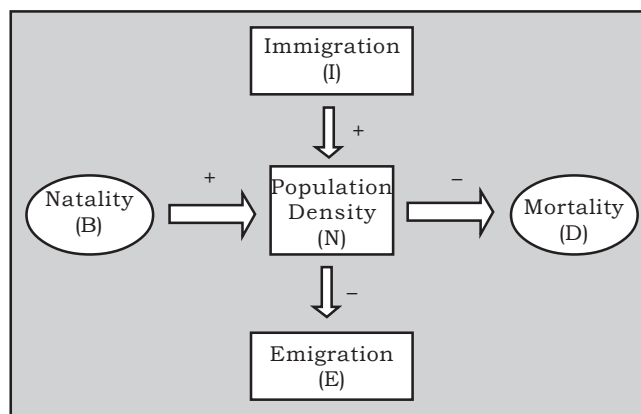
8.7.2. Population Growth

The size of a population for any species is not a static parameter. It keeps changing in time, depending on various factors including availability of food, 'predation pressure' and '**environmental resistance**'. These changes in population density provide information whether it is '*flourishing* or *declining*'. Whatever might be the ultimate reasons, the density of a population in a given habitat during a given period, fluctuates due to changes in four basic processes/ factors. **Natality** and **immigration** contribute to an increase in population density where as **mortality** and **emigration** lead to a decrease.

- (i) **Natality** refers to the number of births during a given period in the population that are added to the initial number of individuals.
- (ii) **Mortality** is the number of deaths in the population during a given period.
- (iii) **Immigration** is the number of individuals of the same species that have come into the habitat/ population from elsewhere/ another population during the time period under consideration.
- (iv) **Emigration** is the number of individuals of the population who left the habitat / population, during the time period under consideration.

So, if "**N**" is the population density at a given time '**t**', its density at the time **t + 1** is $N_{t+1} = N_t + \{(B + I) - (D + E)\}$

It can be seen from the above equation that population density will increase if the number of births plus the number of immigrants (**B + I**) is more than the number of deaths plus the number of emigrants (**D + E**), otherwise it will decrease. Under normal conditions, births and deaths are the most important factors influencing population density, the other two factors assuming importance only under special conditions. For instance, if a new habitat is just being colonised, immigration may contribute more significantly to the population growth than the factors affecting population density birth rate.



Growth Models

Does the growth of a population with time show any specific and predictable pattern? Uncontrolled growth of human population and resultant problems in our country are seriously concerning our planners. Therefore naturally we are curious to know whether different animal populations in nature behave the same way or

show some restraints on growth. Perhaps we can learn a lesson or two from nature on how to control population growth.

- (i) **Exponential growth:** Resource (**food** and **space**) availability is essential for the unhindered growth of a population. Ideally, when resources in the habitat are unlimited (environmental resistance is almost nil), each species has the potential ability to grow in number, as Darwin observed while developing his theory of Natural Selection. In such conditions, the population grows in an '**exponential**' or '**geometric**' fashion. If in a population of size **N**, the birth rates (not total number but *per capita* births) are represented as **b** and death rates (again, *per capita* death rates) as **d**, then the increase or decrease in **N** (**dN/dt**) during a unit time period **t** will be:

$$\frac{dN}{dt} = (b - d)N$$

Let

$$(b - d) = r, \text{ then}$$

$$\frac{dN}{dt} = rN$$

The '**r**' in this equation is called the '**intrinsic rate of natural increase**' and is a very important parameter chosen for assessing impacts of any biotic or abiotic factor on population growth.

To give you some idea about the range of **r** values, for the Norway rat, the **r** is 0.015, and for the flour beetle- it is 0.12. In 1981, the **r** value for human population in India was **0.0205**. To calculate 'r', we need to know the rates of birth and death.

The above equation describes the exponential or geometric growth pattern of a population (see figure) and results in a J-shaped curve when we plot **N** in relation to time. Using the principles of calculus, we can derive the integral form of the exponential growth equation as

$$N_t = N_0 e^{rt} \quad \text{where}$$

N_t = Population density after the time **t** ;

N₀ = Population density at the time zero (initial time);

r = Intrinsic rate of natural increase

e = The base of natural logarithms (2.71828)

Any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time. Darwin showed how even a slow growing animal like elephant could reach enormous numbers in the absence of '**natural checks**'.

- (ii) **Logistic growth:** No population of any species in nature has at its disposal unlimited resources to permit exponential growth. This leads to competition between individuals for limited resources. Eventually, the 'fittest' individual will

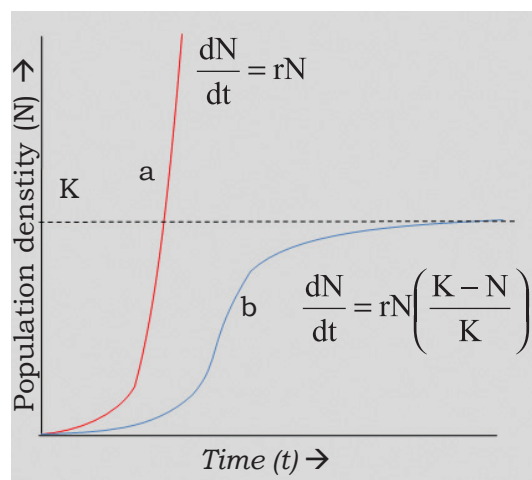
survive and reproduce. The governments of many countries have also realised this fact and introduced various control measures to limit growth of human population. In nature, any given habitat will have enough resources to support a maximum possible number, beyond which no further growth is possible. Let us call this limit as nature's **carrying capacity (K)** for that species in that habitat.

A population growing in a habitat with limited resources show initially a **lag phase**, followed by phases of **acceleration** and **deceleration** and finally an **asymptote/ stability**, when the population density reaches the carrying capacity. A plot of **N** in relation to time (**t**) results in a **sigmoid curve**. This type of population growth is called **Verhulst-Pearl Logistic Growth** (see figure) and is described by the following equation:

$$\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$$

Where

- N** = Population density at the time t
r = Intrinsic rate of natural increase
K = Carrying capacity



Population growth curve

- 'a' : when resources are not limiting the growth plot is exponential.
 'b' : when resources are limiting the growth. plot is logistic,
 'K' : is carrying capacity

Since resources for growth for most animal populations are fixed and become limiting sooner or later, the logistic growth model is considered a more realistic one.

Population Growth curves: Curve 'a' shows the *exponential growth* in population when resources are unlimited; Curve 'b' shows the *logistic growth* in population when resources are a limiting factor where **K** is the *carrying capacity*.

8.7.3 Life history variation

Populations tend to maximise their reproductive fitness in their habitats. This is also called Darwinian fitness (high '**r**' value). Under a particular set of selection pressures, organisms evolve towards the most efficient reproductive strategy. Some organisms breed only once in their lifetime (e.g. Pacific salmon fish, bamboo plant) while others breed many times during their lifetime (e.g. most birds and mammals). Some produce a large number of small-sized offspring (e.g. Oysters, pelagic fishes) while others produce a small number of large-sized offspring (e.g. birds, mammals). The question is – which traits are desirable for maximising fitness? Currently this is an important area of research in ecology. Ecologists suggest that life history traits of organisms have evolved in relation to the constraints imposed by the abiotic and biotic components of the habitat in which they live.

8.8 ENVIRONMENTAL ISSUES

- 8.8.1 Air pollution and its control Noise pollution
- 8.8.2 Water pollution and its control
- 8.8.3 Soil pollution
- 8.8.4 Greenhouse effect and Global warming
- 8.8.5 Ozone depletion in the stratosphere
- 8.8.6 Degradation by improper resource utilisation and maintenance

Pollution

With an enormous increase in human population over the last 100 years, there is a heavy demand for food, water, clothing, home, roads, transport facilities and other innumerable commodities needed by man. To meet the demand man is exerting tremendous pressure on natural resources, thereby causing **environmental pollution**.

Pollution is an undesirable change in the physical, chemical or biological characteristics of the environment due to natural causes and human activities. Any deviation from the natural composition of the environment which causes adverse effects is also described as pollution. The agents which cause pollution are called **pollutants**.

Types of pollution

More commonly pollution is classified according to the type of environment that is affected. Mainly there are three types. 1. Air Pollution 2. Water Pollution and 3. Soil Pollution.

8.8.1 Air Pollution and its control

Earth is surrounded by an envelope of air consisting of various gases which is called atmosphere. As a blanket of gases atmosphere acts as a thermal insulator and regulates the temperature on the earth by selectively absorbing U.V. rays of solar radiation. The composition of the main gases in dry air by volume is Nitrogen **78.09%**, Oxygen **20.94%**, Argon **0.93%**, Carbon dioxide **0.03%**. Life cannot exist on earth without oxygen. Air pollutants cause injury to all living organisms. They reduce growth and yield of crops. They are harmful to the respiratory system of humans and animals. Increase in the concentration of pollutants or duration of exposure increase the harmful effects on the organisms.

The major air pollutants

1. Carbon monoxide (CO)

It is produced mainly due to incomplete combustion of fossil fuels. Automobiles are a major cause of CO pollution in larger cities and towns. Automobile exhausts, fumes from factories, emissions from power plants, forest fires and even burning of fire-wood contribute to CO pollution. Haemoglobin has greater affinity for CO and so CO competitively interferes with oxygen transport. CO causes symptoms such as **headache** and **blurred vision** at lower concentrations. In higher concentrations, it leads to coma and death.

2. Carbon Dioxide (CO₂)

Carbon dioxide is the main pollutant that is leading to **global warming**. Plants utilize CO₂ for photosynthesis and all living organisms emit carbon dioxide in the process of respiration. With rapid urbanization, automobiles, aeroplanes, power plants, and other human activities that involve the burning of fossil fuels such as gasoline, carbon dioxide is turning out to be an important pollutant of concern.

3. Sulphur Dioxide (SO₂)

It is mainly produced by burning of fossil fuels. Melting of sulphur ores is another important source for SO₂ pollution. Metal smelting and other industrial processes also contribute to SO₂ pollution. Sulphur dioxide and nitrogen oxides are the major causes of **acid rains**, which cause acidification of soils, lakes and streams, and also accelerated corrosion of buildings and monuments. High concentrations of sulphur dioxide (SO₂) can result in breathing problems in asthmatic children and adults. Other effects associated with long-term exposure to sulphur dioxide, include respiratory illness, alterations in the lungs' defenses and aggravation of existing cardiovascular problems.

To control SO₂ pollution, the emissions are filtered through scrubbers*. Scrubbers are devices that are used to clean the impurities in exhaust gases .

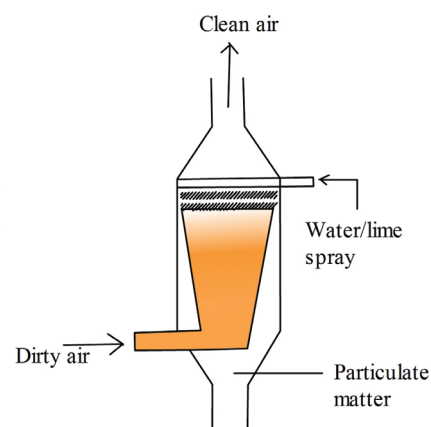


Fig. 8.22 Scrubber

4. Nitrogen Oxides

Nitrogen oxides are considered to be major primary pollutants. The source is mainly automobile exhaust. The air polluted by nitrogen oxides is not only harmful to humans and animals, but also dangerous for the life of plants. Nitrogen oxide pollution also results in **acid rains** and formation of **photochemical smog**. The effect of nitrogen oxides on plants include the occurrence of necrotic spots on the surface of leaves. Photosynthesis is affected in crop plants and the yield is reduced. Nitrogen oxides combine with volatile organic compounds by the action of sunlight to form secondary pollutants called **Peroxyacetyl nitrate** (PAN) which are found especially in photochemical smog*. They are powerful irritants to eyes and respiratory tract.

* Photochemical smog is a type of air pollution caused by reactions between sunlight and pollutants such as hydrocarbons and nitrogen dioxide. Photochemical smog can be extremely harmful, leading to irritations of the respiratory tract (respiratory illness) and eyes.

*“ Scrubbers’ are air pollution control devices used to remove some particulates or gases from industrial exhausts. The term is used to describe systems that inject a dry reagent into a polluted exhaust stream to ‘wash out’/ ‘neutralise’ acid gases. Lime stone slurry is often used in scrubbers (slurry means a thin mixture of insoluble substances such as cement, dust, etc., with water)

5. Particulate matter/Aerosols

Tiny particles of solid matter suspended in a gas or liquid constitute the 'particulate matter'. 'Aerosols' refer to particles and/or liquid droplets and the gas together (**a system of colloidal particles dispersed in a gas***). Combustion of 'fossil fuels' (petrol, diesel, etc.), fly ash produced in thermal plants, forest fires, cement factories, asbestos mining and manufacturing units, spinning and ginning mills etc., are the main sources of particulate matter pollution. According to the **Central Pollution Control Board (CPCB)** particles of 2.5 micrometers or less in diameter are highly harmful to man and other air breathing organisms.

* These microscopic solids or liquid droplets that are so small and can get deep into the lungs cause decreased lung function, aggravated asthma, premature death in people with heart or lung disease, development of chronic bronchitis; irregular heartbeat, etc.

Electrostatic precipitator is a widely used 'filter' for removing particulate matter

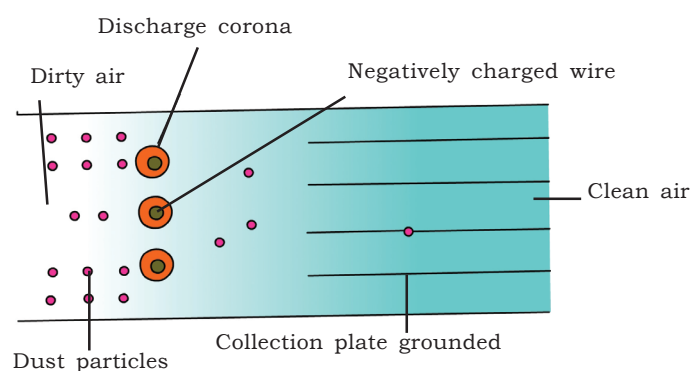


Fig 8.23 Electrostatic precipitator

from the exhaust of thermal power plants. It can remove 99% particulate matter. It has high voltage electrodes which produce a 'corona' that releases electrons. These electrons attach to dust particles making them negatively charged. These are collected by **collecting plates** which attract the charged particles. The air flowing between the plates is kept in low velocity so as to allow the dust particles to settle on plates. Thus clean air is released into the atmosphere.

Measures to control Air Pollution caused by automobiles

1. Proper maintenance of automobiles along with use of diesel or unleaded petrol.
2. Fitting '**catalytic converters**' to the automobiles having expensive metals namely platinum, palladium and rhodium as catalysts which reduce emission of poisonous gases.

Noise Pollution

In India, **Air (Prevention and control of pollution) Act** came into force in 1981. In 1987 it was amended to include noise also as an air pollutant. Undesirably high sounds constitute **noise pollution**. Sound is measured in units called decibels (dB). The human ear is sensitive to sounds ranging from **0 to 180 dB**. **0 dB** is **threshold**

limit of hearing and **120 dB** is threshold limit for sensation of pain in the ear. Any noise above **120 dB** is considered to be a noise pollution. A brief exposure to extremely high sound level, (**150 dB** or more generated by jet planes while taking-off) may damage eardrums causing permanent hearing impairment. Even long term exposure to a relatively lower level of noise of cities may also cause hearing impairment. Noise also causes auditory fatigue, anxiety, sleeplessness (**insomnia**), increased heart beat, altered breathing pattern thus causing considerable stress to humans.

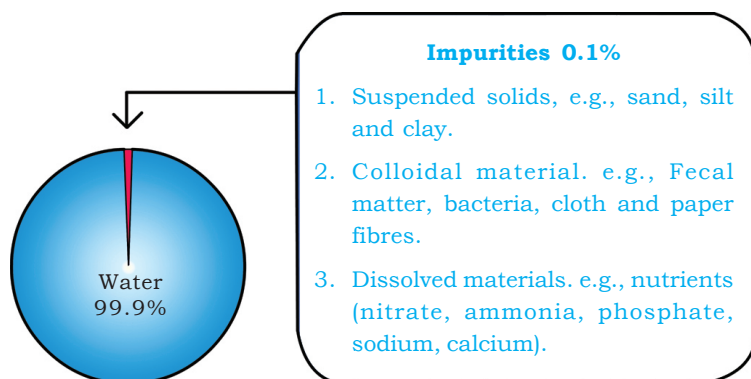
8.8.1.1 Controlling vehicular air pollution: A case study

Delhi leads the country not only as the National Capital but also in its level of air pollution. Air pollution problem in the city of Delhi reached such a serious stage that a public interest litigation (court case) was filed in the Supreme Court of India. Under the directions of the Apex Court, the Government took various measures to control air pollution. The entire fleet of public transport buses were switched over to using compressed natural gas (**CNG**) as 'fuel' instead of diesel by the end of 2002. CNG is cheaper than petrol or diesel. It is the most efficient fuel and burns almost completely or very little of it is left un-burnt in automobiles. Simultaneously parallel steps such as phasing out of old vehicles, using **unleaded** and **low sulphur petrol** and diesel, using catalytic converters in vehicles, application of stringent pollution level norms for vehicles etc., were taken by the Government. The efforts paid well and the air quality of Delhi has significantly improved.

The Government of India, through a new **Auto Fuel Policy** has laid out a roadmap to cut down vehicular pollution in Indian cities. More stringent norms for fuels are specified to steadily reduce the sulphur and the content of aromatics in petrol and diesel fuels. Euro -II norms, for example, stipulate that sulphur be controlled at **350 ppm** (parts-per-million) in diesel, and **150 ppm** in petrol. Aromatic hydrocarbons are to be contained (limited) at 42 per cent of the concerned fuel. The goal, according to the roadmap, is to reduce sulphur to **50 ppm** in petrol and diesel and bring down the level of the aromatic hydrocarbons to 35 per cent. Correspondingly/concurrently, engines of vehicles also are to be upgraded. The **Bharat Stage - III** (equivalent to **Euro-III norms**) is currently in place in many cities such as Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Pune, Surat, Kanpur and Agra, etc. All automobiles in these major cities should have met **Euro-IV norms** by April 1, 2010.

8.8.2 Water pollution and its control

Ancient civilizations flourished along river banks. Primitive life evolved in water only. The bodies of organisms contain water to the extent of nearly 90% of their body weight. Sea water constitutes 97% of water available on the planet. Fresh water which is required for our utility constitutes only about 3%.



Inferior quality of water, caused by pollution of natural waters is a major problem world is facing today. It is posing a serious threat in developing countries and India, of course, is not an exception. Many water bodies in general and almost all the rivers in India are grossly polluted either by sewage or discharge of industrial effluents.

8.8.2.1 Ganga Action Plan

The holy river Ganga along with its 2500 kms. stretch, has been receiving for decades, millions of gallons of untreated domestic sewage from many cities and towns and also toxic industrial effluents from thousands of industries. As a result, it is becoming a **stinking dump** of various toxic substances. The water is becoming unfit for drinking and cultivation. Kanpur, Allahabad, Varanasi, Patna, and Kolkata are identified to be the main centres for causing pollution of rivers. Government of India launched a plan in 1985 with the aim of purifying the waters of river Ganga (**Ganga Action Plan**). Due to change of scope of river pollution control works, the Govt. of India renamed the programme as **National River Conservation Project / Plan**, instead of Ganga Action Plan, which is now spread in 165 towns of 17 different states.

The major water pollutants

1. Domestic Sewage

Sewage is the major source of water pollution in large cities and towns. It mainly consists of human and animal excreta and other waste materials. It is usually released into freshwater bodies or sea directly. As per the regulations the sewage has to be passed through **treatment plants** before it is released into the water courses. Only 0.1 percent of impurities from domestic sewage are making these water sources unfit for human consumption. In the treatment of sewage, solids are easy to remove.

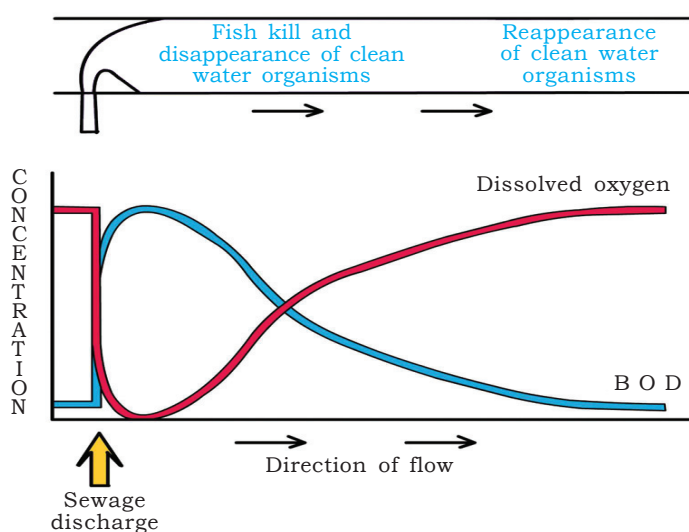


Fig. 8.23 Effect of sewage discharge on some important characteristics of a river

Removal of dissolved salts such as nitrates, phosphates and other nutrients and toxic metal ions and organic compounds is much more difficult. Domestic sewage primarily contains biodegradable organic matter, which will be readily decomposed by the action of bacteria and other microorganisms.

Biological Oxygen Demand (BOD)

BOD is a measure of the content of biologically degradable substances in sewage. The organic degradable substances are broken down by microorganisms using oxygen. The demand of oxygen is measured in terms of the oxygen consumed by microorganisms over a period of 5 days (**BOD 5**) or seven days (**BOD 7**). BOD forms an index for measuring **pollution load** in the sewage. Microorganisms involved in biodegradation of organic matter in water bodies consume a lot of oxygen, and as a result there is a sharp decline in dissolved oxygen causing death of fish and other aquatic animals.

Algal blooms

Presence of large amounts of nutrients in waters also causes excessive growth of planktonic algae and the phenomenon is commonly called 'algal blooms'. Algal blooms impart distinct colour to the water bodies and reduce the quality of water. It also causes mortality of fish. Some algae which are involved in algal blooms are toxic to human beings and animals.

Excessive growth of aquatic plants such as the *common water hyacinth* (*Eichhornia crassipes*), the world's most problematic aquatic weed which is also called '**Terror of Bengal**' causes blocks in our water ways. They grow faster than our ability to remove them. They grow abundantly in **eutrophic** water bodies (water bodies rich in nutrients) and lead to imbalance in the ecosystem dynamics of the water body.

Sewage arising from homes and hospitals may contain undesirable pathogenic microorganisms. If it is released untreated into water courses, there is a likelihood of outbreak of serious diseases, such as dysentery, typhoid, jaundice, cholera etc.,

2. Industrial Effluents

Untreated industrial effluents released into water bodies pollute most of the rivers, fresh water streams, etc., Effluents contain a wide variety of both inorganic and organic pollutants such as oils, greases, plastics, metallic wastes, suspended solids and toxins. Most of them are non-degradable. Arsenic, Cadmium, Copper, Chromium, Mercury, Zinc, and Nickel are the common heavy metals discharged from industries.

EFFECTS: Organic substances present in the water deplete the dissolved oxygen content in water by increasing the **BOD**(Biological oxygen demand) and **COD**(Chemical oxygen

demand). Most of the inorganic substances make the water unfit for drinking. Outbreaks of dysentery, typhoid, jaundice, cholera etc., are caused by sewage pollution.

Biomagnification

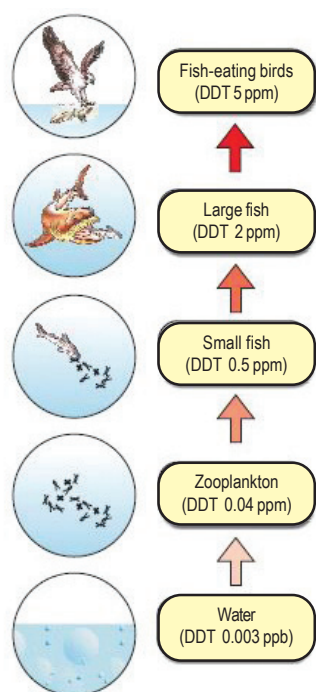


Fig. 8.24 Biomagnification of DDT in an aquatic food chain

Increase in the concentration of the pollutant or toxicant at successive trophic levels in an aquatic food chain is called **Biological Magnification** or **Bio-magnification**. This happens in the instances where a toxic substance accumulated by an organism is not metabolized or excreted and thus passes on to the next higher trophic level. This phenomenon is well known regarding DDT and mercury pollution.

As shown in the Fig. 8.24, the concentration of **DDT** is increased at successive trophic levels starting at a very low concentration of **0.003 ppb** (ppb = parts per billion) in water, which increased to a high concentration of **5 ppm** (ppm = parts per million) in fish-eating birds, through bio-magnification (It can reach maximum upto 25 ppm). High concentrations of DDT disturb **calcium metabolism in birds**, which causes thinning of egg shell and their premature breaking, eventually causing decline in bird populations.

Eutrophication

Natural **ageing** of a lake by nutrient enrichment of its water is known as **eutrophication**. In a young lake, the water is cold and clear, supporting little life. Gradually nutrients such as nitrates and phosphates are carried into the lake via streams, in course of time. This encourages the growth of aquatic algae and other plants. Consequently the animal life proliferates, and organic matter gets deposited on the bottom of the lake. Over centuries, as silt and organic debris piles up, the lake grows shallower and warmer. As a result, the aquatic organisms thriving in the cold environment are gradually replaced by warm-water organisms. Marsh plants appear by taking root in the shallow regions of the lake. Eventually, the lake gives way to large masses of floating plants (bog) and finally converted into land.

A **bog** is a wet, spongy, acidic ground composed of decayed vegetable matter.

Depending upon the climatic conditions, size of the lake and other factors, the natural ageing of a lake may span thousands of years. However, pollutants from human activity (**anthropogenic**) radically accelerate the aging process. This phenomenon is called '**Cultural** or **Accelerated eutrophication**'.

During the past century, lakes in many parts of the earth have been severely eutrophied by sewage, agricultural and industrial wastes. The prime contaminants

are nitrates and phosphates, which are the '**chief plant nutrients**'. The dissolved oxygen which is vital to other aquatic life is depleted. At the same time, other pollutants flowing into the lake may poison the whole population of fish, whose decomposing remains further deplete the dissolved oxygen content in the water.

Thermal pollution

Water is used as a coolant in Thermal power plants and other industries. Hot water flowing out of industries also constitute an important category of pollutants. Thermal waste water eliminates sensitive organisms (**stenothermal** organisms such as fish- especially the juveniles) downstream and may enhance the growth of plants and fish in extremely cold areas but, only after causing damage to the indigenous flora and fauna.

8.8.2.2 A Case Study Of Integrated Waste Water Treatment

Wastewater including sewage can be treated in an integrated manner, by combining both artificial and natural processes. A good example for such an initiative comes from the town of **Arcata**, situated along the northern coast of **California**. In collaboration with biologists from the Humboldt State University, the people of the town created an integrated waste water treatment process within a natural system. The cleaning of waste water is done in two stages.

1. In the **first stage**, the conventional sedimentation, filtering and chlorination treatments are given. Even after this treatment, the waste water contains lots of harmful pollutants such as dissolved heavy metals.
2. In the **second stage**, the water is discharged into a series of six connected marshes over 60 hectares of marshland seeded with appropriate plants, algae, fungi and bacteria as suggested by biologists. These flora of the marshland **neutralise, absorb and assimilate the pollutants**. As the water flows through the marshes, it is purified in a natural way and can be used for cultivation or other purposes.

The marshes also constitute a **sanctuary**, with a high level of **biodiversity** in the form of fishes, animals and birds that inhabit the area. A citizens' group called **Friends of the Arcata Marsh** (FOAM) are responsible for the upkeep and safeguarding of the wonderful project.

Ecological Sanitation - 'Ecosan Toilets'

Generally it is assumed that removal of wastes requires water, which means **creation of sewage**. If water is not used to dispose off human waste like excreta, and if one didn't have to flush the toilet after its use, a large amount of water can be saved. This is already a reality. Ecological sanitation is a sustainable system for handling human excreta, using '**Dry composting toilets**'. This is a practical, hygienic, efficient and cost-effective solution to human waste disposal. The key point to note here is that,

with this composting method, human excreta can be recycled into a resource (as natural fertiliser), which reduces the need for chemical fertilizers. 'EcoSan' toilets are in use in many parts of Kerala and Sri Lanka.

8.8.3 Soil Pollution (Land Pollution)

Soil or land pollution is mainly due to solid wastes and agro-chemicals such as fertilizers and pesticides.

Solid Wastes

Any thing (substance/material/articles/goods) that is thrown out as waste in solid form is referred to as **solid waste**. Municipal solid wastes are wastes from homes, offices, institutions, shops, hotels, restaurants etc., in towns and cities.

The municipal solid wastes generally consist of paper, food wastes, plastics, glass, metals, rubber, leather, textile, etc., The wastes are **burnt to reduce the volume** of the wastes. But generally wastes are not completely burnt and left as open dumps which often serve as the breeding grounds for rats and flies. As the substitute for open-burning dumps, **sanitary landfills** are adopted. In a sanitary landfill, wastes are dumped in a depression or trench after compaction, and covered with dirt everyday. There is a danger of seepage of chemicals and pollutants from these landfills, which may contaminate the underground water resources.

The best solution is to develop awareness in the society on these environmental issues. All wastes that we generate can be categorized into three types (a) **bio-degradable**, (b) **recyclable** and (c) **non-biodegradable**. It is important that all garbage generated should be sorted out category wise. The reusable or recyclable material has to be separated out and utilised. (Rag-pickers in the streets are doing a great job of separation of materials for recycling.) The biodegradable materials can be put into deep pits in the ground and be left for natural breakdown. The remaining non-biodegradable waste left over is to be disposed off properly.

The prime goal should be to reduce our garbage generation. But we are increasing the use of non-biodegradable products. We are packaging products of our daily use such as milk and water also in polythene bags. In cities and towns, many purchased things are packed in **polystyrene** and **plastic** packets. Thus we are contributing heavily to environmental pollution. State Governments across the country are trying to educate people on the reduction in use of plastics and use of **eco-friendly packaging**. We can do our bit by using carry-bags made of cloth or other natural fibres when we go for shopping and by refusing polythene bags.

i) Hospital Wastes

Hospitals generate hazardous wastes that contain disinfectants, harmful chemicals and also pathogenic micro-organisms. Such wastes also require careful treatment and

disposal. The use of **incinerators** (to burn wastes) is essential for disposal of hospital waste.

ii) Electronic wastes (e-wastes):

Irreparable computers and other electronic goods constitute the modern day pollutants called electronic wastes (e-wastes). E-wastes are buried in landfills or incinerated. Over half of the e-wastes generated in the developed world are exported to developing countries, mainly to China, India and Pakistan, where metals like copper, iron, silicon, nickel and gold are recovered during recycling process. Unlike developed countries, which have specifically built facilities for recycling of e-wastes, recycling in developing countries often involves manual participation thus exposing workers to toxic substances present in e-wastes. Eventually recycling is the only solution for the treatment of e-wastes provided it is carried out in an **environmental friendly** manner.

8.8.3.1 A Case Study Of Remedy For Plastic Waste

A Bangalore based plastic sack manufacturer **Ahmed Khan**, found an ideal solution to the ever-increasing problem of accumulating plastic waste. He is in the field of producing plastic sacks for more than 20 years. A few years ago, he realised the threat posed by plastic waste. As a remedy, his company developed a fine powder of recycled plastic mixture named **polyblend**. Polyblend is mixed with the **bitumen** that is used to lay roads. In collaboration with R.V.College of Engineering and the Bangalore City Corporation, Ahmed Khan proved that the life of roads laid with **polyblend** and **bitumen** mixture is tripled. It is observed that polyblend enhanced the bitumen's water repellant properties, and helped increase its road life. The raw material for creating polyblend is any plastic film waste. The cost of plastic waste given to rag pickers rose from Rs. 0.40 per kg to Rs.6 as was offered by Khan. Using Khan's technique, several roads have already been laid in Bangalore. Khan's company will soon be running short of plastic waste in Bangalore, to produce Polyblend. Thanks to innovations like Polyblend, we could find at least one way to overcome the problem of plastic wastes.

iii) Agro-chemicals and Their Effects

In the wake of the **Green Revolution**, use of inorganic fertilisers and pesticides has increased many times, for enhancing crop production. Pesticides, herbicides, fungicides, etc., are being increasingly used. They are also toxic to non-target organisms such as earthworms, nitrogen fixing bacteria, etc., that are important components of soil eco-system. Moreover due to bio-magnification, the harmful chemicals pose a great threat to human health. Indiscriminate use of fertilizers will lead to increased drain of nutrients into the nearby aquatic ecosystems causing eutrophication and the consequent effects.

8.8.3.2 Case Study of organic farming

Integrated organic farming is a **zero-waste** procedure, where recycling of waste products is efficiently carried out. Wastes originated from one process are used as nutrients for other processes. This allows the maximum utilisation of resource and increases the efficiency of production. A method practiced by **Ramesh Chandra Dagar**, a farmer in Sonipat, Haryana, is a very good example for this. He integrated bee-keeping, dairy management, water harvesting, composting and agriculture in a chain of processes. All these processes support one another and allow an extremely economical and sustainable venture. Crop waste and cattle excreta (dung) are used to create **compost**, which can be used as a natural fertilizer. Natural **bio-gas** generated in the process can be used for meeting the energy needs of the farm. Enthusiastic about spreading information and helping in the practice of integrated organic farming, Dagar has created the **Haryana Kisan Welfare Club**.

iv) Radioactive wastes

Initially, nuclear energy was considered a non-polluting way for generating electricity. Later on, it was realised that the use of nuclear energy has two very serious inherent problems. The first is **accidental leakages**, as occurred in the **Three Mile Island (USA)** and **Chernobyl (Russia)** and the second is the **safe disposal** of radioactive wastes.

Radiation, that is released from nuclear waste is extremely dangerous to biological organisms, because it induces mutations. Exposure to high doses of nuclear radiation can lead to cancers (e.g. leukemia)., Therefore, nuclear waste is an extremely **potent pollutant** and has to be dealt with utmost caution. Storage of nuclear wastes should be done in suitably shielded containers and buried deep in the soil or oceans (about 500 meters). Even when done so, geological upheavals can bring them up, some day and cause radiation.

8.8.4 Green house effect and global warming

The term '**Greenhouse effect**' has been derived from a phenomenon that occurs in a 'greenhouse'. Greenhouse is a small glass house and is used for growing plants especially during winter. In a greenhouse the glass panel allows the passage of light into it, but does not allow heat to escape (as it is reflected back). Therefore, the greenhouse warms up, very much like inside a car that has been parked in the sun for a few hours.

The greenhouse effect is a naturally occurring phenomenon that is responsible for heating of the Earth's surface and atmosphere. It would be surprising to know that without greenhouse effect the average temperature of the Earth's surface would have been a chilly -18°C rather than the present average of 15°C .

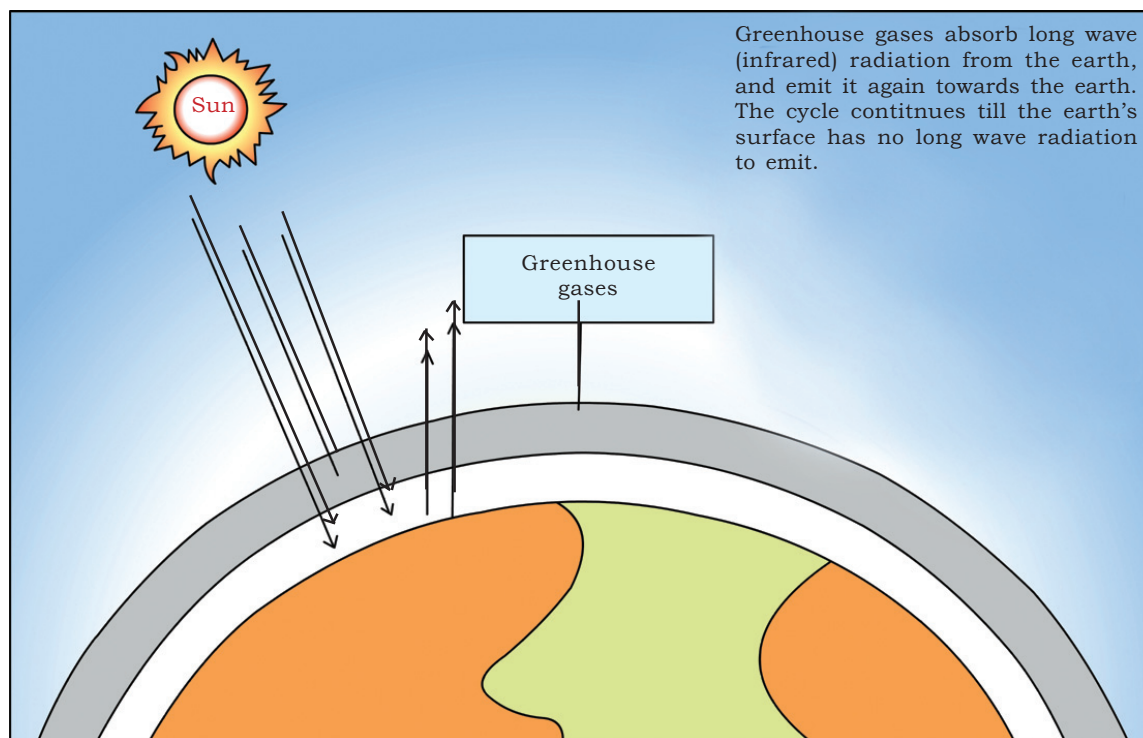


Fig. 8.25 Sunlight energy at the outermost atmosphere

When sunlight reaches the outermost layer of the atmosphere, clouds and gases reflect about one-fourth of the incoming solar radiation, and absorb some of it. Almost half of the incoming solar radiation falls on the Earth's surface and heats it up, while a small proportion is reflected back. The Earth's surface reflects heat in the form of infrared radiation but part of this does not escape into space as atmospheric gases (e.g., carbon dioxide, methane, etc.) absorb a major fraction of it. The molecules of these gases radiate heat energy, and a major part of which again comes back to the Earth's surface, thus **heating it up once again**. The above-mentioned gases – **carbon dioxide** and **methane** are commonly known as **greenhouse gases**, because they are responsible for the **greenhouse effect**.

Increase in the level of greenhouse gases has led to considerable heating of the Earth leading to **global warming**. During the past century, the temperature of the Earth has increased by 0.6 °C, most of it during the last three decades. Scientists believe that this rise in temperature is leading to severe changes in the environment. Global warming is causing climatic changes (e.g. such as **El Nino effect***) and is also responsible for the melting of polar ice caps and other snow caps of mountains such as the Himalayas. Over many years, this will result in a rise in sea levels, all over the world, that can submerge many coastal areas in the world. The total spectrum of

* El nino is unusual weather phenomena caused due to rise in temperature of the surface waters of the Pacific ocean in different parts of the world. Its effects are experienced all over the world and more so in the South American countries

changes that global warming can bring about is a subject that is still under active research.

Global warming – Control measures

1. The measures include cutting down use of fossil fuels.
2. Improving efficiency of energy usage.
3. Planting of trees, and avoiding deforestation.
4. Slowing down the growth of human population.

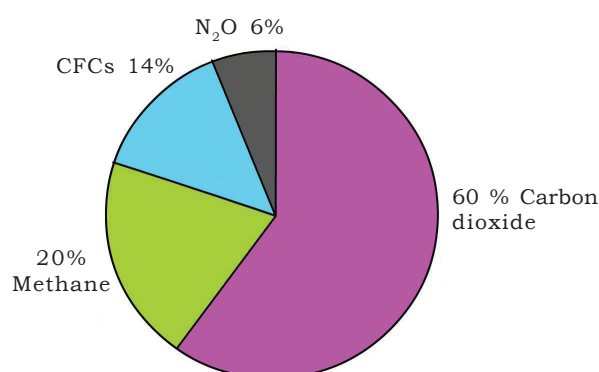


Fig 8.26 Relative contribution of various green house gases to total global warming

Kyoto Protocol

International initiatives are also being taken to reduce the emission of greenhouse gases into the atmosphere. To save earth from the dangers of 'global warming', an international conference was held at Kyoto, Japan (1997). As of September 2011, 191 countries have signed and ratified the protocol. As per this resolution it is decided to achieve the goal of reducing **green house emissions** into the atmosphere by 2008-12 to the levels of 1990.

8.8.5 Ozone depletion in the

stratosphere

Ozone, formed in the lower atmosphere (troposphere) is known as '**bad ozone**' which harms plants and animals. Ozone found in the upper part of the atmosphere called the stratosphere is known as '**good ozone**', and it acts as an efficient **shield** by absorbing ultraviolet radiation from the sun. UV rays are highly injurious to living organisms. DNA and proteins of living organisms preferentially absorb UV radiation, and its high energy breaks the chemical bonds within these molecules. The thickness/ columnar density of the ozone in a **column of air** is measured in terms of **Dobson units** (DU). Ozone gas is continuously formed by the action of UV rays on molecular oxygen and is unstable. It is degraded into molecular oxygen in the stratosphere. There will be generally a balance between production and degradation of ozone in the stratosphere. Of late, the balance has been disrupted due to enhancement of ozone degradation by chlorofluorocarbons (**CFCs**). CFCs are widely used as refrigerants and coolants. CFCs discharged into the lower strata of the atmosphere, move upward and reach the stratosphere. In the stratosphere, Cl atoms are released from CFC molecules by the action of UV rays. These Cl atoms act as catalysts and degrade ozone molecules and release **molecular oxygen**. As the Cl atoms are not consumed in this reaction, they continue to degrade ozone molecules. Hence, whatever CFCs are added to the stratosphere, they have permanent and continuing effects on the levels of ozone. The

depletion of ozone is particularly marked over the **Antarctic region**. This has resulted in the formation of a large area of thinned ozone layer, commonly called as the '**ozone hole**' (Figure).

UV radiation with wavelengths shorter than that of UV-B, are almost completely absorbed by Earth's atmosphere, provided that the ozone layer is intact. But **UV-B damages DNA and may induce mutations**. It causes ageing of skin, damage to skin cells resulting in **skin cancers**. In human eye, cornea absorbs **UV-B** radiation, and a high dose of **UV-B** causes inflammation of **cornea**, called **snow-blindness** formation of cataract, etc., Such exposure may permanently damage the cornea.

Montreal Protocol

Understanding the deleterious effects of depletion of ozone, an international treaty, known as the **Montreal Protocol**, was signed at Montreal (Canada) in **1987** (became **effective in 1989**) to control the emission of pollutants/ substances responsible for ozone depletion. Subsequently many more efforts have been made and protocols have laid down definite roadmaps, separately for developed and developing countries, for reducing the emission of CFCs and other ozone depleting chemicals.

8.8.6 Degradation by improper resource utilisation and maintenance

The degradation of natural resources can occur, not just by the action of pollutants but also by improper resource utilisation practices.

- i) **Soil erosion and desertification**: The development of the fertile **top-soil** takes centuries. But, it can be removed very easily due to human activities like over-cultivation, unrestricted grazing, deforestation and poor irrigation practices, resulting in the formation of arid patches of land. When large barren patches extend and meet over time, a desert is created. Internationally, it has been recognised that **desertification** is a major problem nowadays, particularly due to increased urbanisation.
- ii) **Water-logging and soil salinity**: Irrigation without proper drainage of water leads to water-logging in the soil. Besides affecting the crops, water-logging draws salt to the surface of the soil (**salinisation of the top soil**). The salt then is deposited as a thin crust on the land surface or starts collecting at the roots of the plants.

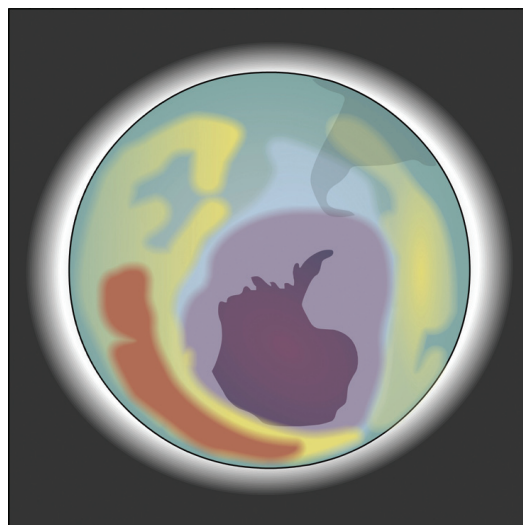


Fig. 8.27 Ozone hole is the area above Antarctica, shown in purple colour, where the ozone layer is the thinnest. Ozone thickness is given in dobson units (see carefully the scale shown in colour violet or red). The ozone hole over Antarctica develops each year between late August and early October. *Curtesy: NASA*

This increased salt content is inimical (unfavourable) to the growth of crops and is extremely damaging to agriculture. Water-logging and soil salinity are some of the problems that have come in the wake of the Green Revolution.

iii) Deforestation: Deforestation is the conversion of forested areas to non-forested ones. According to an estimate, almost forty per cent forests have been lost in the tropics, compared to only one per cent in the temperate regions. At the beginning of the twentieth century, forests covered about **30** per cent of the land of India. By the end of the century, it shrunk to **19.4** per cent, whereas the **National Forest Policy (1988)** of India has recommended **33** percent forest cover for the plains and 67 per cent for the hills as ideal or desirable.

A number of human activities contribute to deforestation. One of the major reasons is the conversion of forest to agricultural land so as to feed the growing human population. Trees are axed for timber, firewood, cattle ranching and for several other purposes. **Slash** (cut) and **burn agriculture**, commonly called as **Jhum cultivation** in the north-eastern states of India and **podu** in other parts, has also contributed to deforestation. In slash and burn agriculture, the farmers cut down the trees of the forest and burn the plant remains. The ash serves as a fertiliser and the land is then used for farming or grazing cattle. After cultivation, the area is abandoned for several years to allow its recovery. The farmers then move on to other areas and repeat this process.

Consequences of deforestation

- 1) Enhanced carbon dioxide concentration in the atmosphere because trees that could hold a lot of carbon in their biomass (use CO_2 for photosynthesis) are lost with deforestation.
- 2) Loss of biodiversity due to habitat destruction.
- 3) Disturbed hydrological cycle.
- 4) Soil erosion.
- 5) Desertification in extreme cases.

Reforestation

Reforestation is the process of restoring a forest that once existed but was destroyed at some point of time in the past. Reforestation may occur naturally in a deforested area. However, we can speed it up by planting trees(saplings) with due consideration to biodiversity that existed earlier in that area.

8.8.6.1 Case Study of People's Participation in the Conservation of Forests

Amrita Devi Bishnoi wildlife protection Award

People's participation in 'protecting forests' has a long history in India. In 1731, the king of Jodhpur in Rajasthan asked one of his ministers to arrange wood for constructing a new palace. The minister and workers went to a forest near a village, inhabited by Bishnois, to cut down trees. The **Bishnoi community** is known for its peaceful co-existence with nature. The effort to cut down trees by the king was thwarted by the Bishnois. A Bishnoi woman **Amrita Devi** showed exemplary courage by hugging a tree and daring king's men to cut her first before cutting the tree. The tree mattered much more to her than her own life. Sadly, the king's men did not heed to her pleas, and cut down the tree along with Amrita Devi. Her three daughters and hundreds of other Bishnois followed her, and thus lost their lives in their effort to save trees. Nowhere in history do we find a commitment of this magnitude when human beings sacrificed their lives for the cause of the environment. The Government of India has recently instituted the "**Amrita Devi Bishnoi Wildlife Protection Award**" for individuals or communities from rural areas that have shown extraordinary courage and dedication in protecting wildlife.

The Chipko Movement of Garhwal Himalayas

In 1974, local women showed enormous bravery in protecting trees from the axe of contractors by hugging the trees. People all over the world have acclaimed the **Chipko movement**. Realising the significance of participation by local communities the Government of India in 1980s has introduced the concept of **Joint Forest Management (JFM)** so as to work closely with the local communities for protecting and managing forests. In return for their services to the forest, the communities get the benefit of various forest products (e.g., fruits, gum, rubber, medicine, etc.).

Indian Government passed Environment (Protection) Act in 1986; Water (Prevention and Control of Pollution) Act in 1974.

GLOSSARY

Autecology: ecology of individual species

Basking: Exposing the body to sun light, to gain temperature.

Benthos: It refers to all the attached, creeping or burrowing organisms that inhabit the bottom of rivers, lakes and sea.

Biomass: The total mass of living material within a specified area at a given time.

Blubber: It is a specialized subcutaneous layer of fat found only in marine mammals. It is almost continuous across the body of marine mammals but absent on appendages.

Brackish water: An intermediate zone between freshwater and marine water.

Camouflage Concealing coloration (e.g., melanism) and morphology (e.g. stick insects) as defence against predation

Chemoautotrophs: These comprise bacteria that obtain energy from the oxidation of simple inorganic compounds and can use the energy released to assimilate CO_2 and transfer the energy into organic compounds. E.g. *Thiobacillus* species.

Climate: The climate of an area can be described by its mean values of temperature, rainfall, wind speed.

Community: The total living biotic component of an ecosystem, including plants, animals and microbes.

Competitive exclusion: It is often known as Gause principle.

Cyclomorphosis: Cyclic change in phenotype, such as seasonal changes in morphology, particularly conspicuous among cladoceran crustacean and rotifers.

Detritus: Non living organic matter. Usually refers to particulate matter to that of plant rather than animal origin. E.g. leaf litter.

Diapause: It is a condition of arrested growth or reproductive development common in many organisms, particularly insects, during unfavourable conditions.

Dimictic Lake: The lake that undergoes two periods of complete vertical mixing, usually in the spring and the fall. During the summer dimictic lakes are thermally stratified.

Edaphic factors: The physical, chemical and biotic characteristics of the soil that influence plant growth and distribution.

Estuary: It is a place where river joins the sea. The water in an estuary is subjected to seasonal variations in salinity. The water is called brackish water. The animals living there are euryhaline.

Gemmules: These are the internal buds that appear in the asexual reproduction of sponges. Gemmules are made up of amoebocytes and covered by a layer of spicules, and can survive in unfavourable conditions.

Leaching: Removal of soluble components by flowing water from soil.

Mycorrhizae: The symbiotic association of mycelium of a fungus with the roots of a seed plant.

Osmotrophic nutrition: Intake of pre-digested food material through the body surface.

Pedonic forms: The organisms which depend on substratum (or support) in an aquatic ecosystem.

Periphyton: The communities of tiny organisms like protozoa, insect larvae, snails that live on the surfaces of aquatic plants.

Savannas: Grassland region with scattered trees in subtropical and tropical regions.

Standing crop: The mass of vegetation in a given area at one particular time. Although most of often applied to plant material, the term includes animal biomass.

Denitrificaion: Denitrification is a microbially facilitated process of nitrate reduction that may ultimately produce molecular nitrogen (N_2) through a series of intermediate gaseous nitrogen oxide products.

Decomposer: Decomposers (or saprotrophs) are organisms that break down dead or decaying organisms, and in doing so carry out the natural process of decomposition.

Sedimentary: Formed by the accumulation and consolidation of mineral and organic fragments that have been deposited by water, ice, or wind.

Asymptote: A line which approaches nearer to some curve than assignable distance, but, though infinitely extended, would never meet it.

Chemo-autotroph: An organism (typically a bacterium or a protozoan) that obtains energy through chemical process, which is by the oxidation of electron donating molecules from the environment, rather than by photosynthesis.

Mortality: Death rate or the number of individuals that died in a population in a unit time.

Natality: Birth rate or the number of individuals produced in a population in unit time.

Acid rains: Acid rain is a rain or any other form of precipitation that is unusually acidic.

Algal bloom: An algal bloom is a rapid increase or accumulation in the population of algae (typically microscopic) in an aquatic system.

Biodegradable: Capable of being broken down especially into harmless products by the action of living things (as microorganisms).

Biological Oxygen Demand (BOD): The amount of dissolved oxygen needed by aerobic biological organisms in a body of water to break down organic material present in a given water sample at certain temperature over a specific time period.

Chemical Oxygen Demand (COD): A test procedure based on the chemical decomposition of organic and inorganic contaminants dissolved or suspended in water.

Chloro fluoro carbons (CFC): Any of various halocarbon compounds consisting of carbon, hydrogen, chlorine, and fluorine, once used widely as aerosol propellants and refrigerants. Chlorofluorocarbons are believed to cause depletion of the atmospheric ozone layer.

Deforestation: The removal of a forest or stand of trees where the land is thereafter converted to a non-forest use.

Eutrophication: A process whereby water bodies, such as lakes, estuaries, or slow-moving streams receive excess nutrients that stimulate excessive plant growth (algae, periphyton attached algae, and nuisance plants and weeds).

Fungicides: Substance or preparation, as a spray or dust, used for destroying fungi.

Herbicides: Chemicals used to kill unwanted weeds.

Incinerator: a furnace or apparatus for burning trash, garbage, etc., to ashes.

Land fills: Landfill is a carefully engineered depression in the ground (or built on top of the ground, resembling a football stadium) into which wastes are dumped.

Pesticides: Chemicals used to kill pests, especially insects.

Photochemical smog: A type of air pollution produced when sunlight acts upon motor vehicle exhaust gases to form harmful substances such as ozone (O_3), aldehydes and peroxyacetylnitrate (PAN).

Polyblend: Physical mixture of two or more polymers. Such blends usually yield products with favorable properties of both components.

Scrubber: Scrubbers are commonly used to eliminate potentially harmful dust and pollutants from exhausts. In scrubbers, a liquid, in general water added with active chemicals is sprayed in to the air flow. Aerosol and gaseous pollutants in the air stream are removed by either absorption or chemical reactions with the water solution.

Sewage: Domestic waste water containing various solid and liquid waste materials including human excreta.

Soil erosion: The washing away of soil by the flow of water or wind

Thermal Pollution: Water Pollution caused by hot water coming out from industries, thermal power plants etc., which is harmful to aquatic organisms.

Ultraviolet-B (UV-B): One of the three types of invisible light rays (together with ultraviolet-A and ultraviolet-C) given off by the sun. Although ultraviolet-C is the most dangerous type of ultraviolet light in terms of its potential to harm life on earth, it cannot penetrate earth's protective ozone layer. Therefore, it poses no threat to human, animal or plant life on earth as long as ozone layer is intact. Ultraviolet-A and ultraviolet-B, on the other hand, do penetrate the ozone layer in attenuated form and reach the surface of the planet. UV-A rays cause cells to age and can cause some damage to cells' DNA. They are linked to long-term skin damage such as wrinkles, but are also thought to play a role in some skin cancers. UVB rays can cause direct damage to the DNA, and are the main rays that cause sunburns. They are also believed to cause most skin cancers.

VERY SHORT ANSWER TYPE QUESTIONS

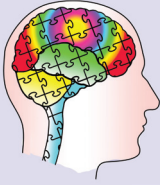
1. Define the term ecology and its branches.
2. Define a community?
3. What is an ecosystem?
4. What is a biome? Name any two biomes you studied.
5. Explain the difference between the 'niche' of an organism and its 'habitat'.
6. How does your body solve the problem of altitude sickness, when you ascend tall mountains?
7. What is the effect of light on body pigmentation?
8. Distinguish the terms phototaxis and photokinesis.
9. What is photoperiodism?
10. Mention the advantages of some UV rays to us.
11. What is cyclomorphosis? Explain its importance in *Daphnia*.
12. What are 'regulators'?
13. What are 'conformers'?
14. Define commensalism. Give one example.
15. Define mutualism. Give one example.
16. What is camouflage? Give its significance.
17. What is Gause's principle? When does it applicable?
18. Distinguish between lotic and lentic habitats.
19. Distinguish between phytoplankton and zooplankton.
20. Distinguish between neuston and nekton.
21. Explain the process of 'leaching'?
22. What is PAR?
23. What is the percentage of PAR, in the incident solar radiation.
24. Explain the terms GPP, NPP.
25. Distinguish between upright and inverted ecological pyramids.
26. Distinguish between primary and secondary productivity.
27. Which air pollutants are chiefly responsible for acid rains?
28. What is BOD?
29. What is biological magnification?
30. Why are incinerators used in hospitals?

SHORT ANSWER TYPE QUESTIONS

1. What is summer stratification? Explain.
2. What is the significance of stratification in lakes?
3. Explain van't Hoff rule.
4. Unlike mammals the reptiles cannot tolerate environmental fluctuations in temperature. How do they adapt to survive in desert conditions?
5. How do terrestrial animals protect themselves from the danger of dehydration of bodies?
6. How do marine animals adapt to hypertonic seawater?
7. Distinguish between euryhaline and stenohaline animals.
8. An orchid plant is growing on the branch of mango tree. How do you describe this interaction between the orchid and mango tree?
9. Predation has a significant role in maintaining of species diversity-discuss.
10. Discuss competitive release.
11. Explain brood parasitism with a suitable example.
12. How do predators act as biological control?
13. Explain the different types of aquatic ecosystems.
14. Explain the terms saprotroph, detritivores, and mineralizers.
15. Define decomposition and describe the process and products of decomposition.
16. What is primary productivity? Give a brief description of the factors that affect primary productivity.
17. Define ecological pyramids and describe with examples, pyramids of numbers and biomass.
18. What are the deleterious effects of depletion of ozone in the stratosphere?
19. Describe 'Green House Effect'?
20. Write critical notes on the following:
 - a) Eutrophication
 - b) Biological magnification
 - c) Ground water depletion and ways for its replenishment

**LONG ANSWER TYPE
QUESTIONS**

1. Write an essay on temperature as an ecological factor.
2. Write an essay on water as an ecological factor.
3. Describe lake as an ecosystem giving examples for the various zones and the biotic components in it.
4. Describe different types of food chains that exist in an ecosystem.
5. Give an account of flow of energy in an ecosystem.
6. List out the major air pollutants and describe their effects on human beings ?
7. What are the causes of water pollution and suggest measures for control of water pollution ?



For ignited minds

- NOT FOR EVALUATION

1. All types of UV rays are not useless to man. How do you defend the statement?
2. Elephant is a large sized animal and an herbivore. How do you justify it?
3. Lakes in temperate countries are much better places for aquatic life than the tropical lakes. Do you agree with above statement? Explain.
4. Even though different kinds of plants live in a lake, there is a sort of 'war' between some species, which is not visible to the naked eye. Justify the comment giving an example.
5. With reference to representation of biomasses in food pyramids, what happens when algae 'bloom' and after a month or two the algae start dying, as they have completed their life cycles.
6. "All photosynthetic organisms are autotrophs, but all autotrophs are not photosynthetic." How do you substantiate this statement.
7. Based on the laws of thermodynamics, guess if bioluminescence influences the 'energy homeostasis' in the body. (Clue: bioluminescence is a catabolic process, in a way.)
8. If the body mass of a tertiary carnivore is 10 KJ, and if the photosynthetic organisms which provide food for it utilize 20% of their produce in their catabolism, what will be the GPP taking into consideration the 10% law.
9. Why do marine birds excrete salts through nasal passages?
10. Can a fungus become a part of the predator chain? If so, can you give an example?

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